

Project 8: Low Frequency Apparatus

Pre-Conceptual Design Report

Case Western Reserve University

R. Mohiuddin, B. Monreal, Y.-H. Sun

Heidelberg University

F. Spanier

University of Illinois, Urbana-Champaign

C.-Y. Liu

Indiana University

M. Oueslati, W. Pettus, A. Reine

Johannes Gutenberg Universität, Mainz

S. Böser, M. Fertl, A. Lindman, C. Matthé, B. Mucoglava, R. Reimann, F. Thomas, L. Thorne

Karlsruhe Institute of Technology, Institute for Astroparticle Physics

Th. Thümmler

Lawrence Berkeley National Laboratory

A. W. P. Poon

Lawrence Livermore National Laboratory

K. Kazkaz

Massachusetts Institute of Technology

J. A. Formaggio, M. Li, J. Peña, J. Stachurska, W. Van De Pontseele

Pacific Northwest National Laboratory

J. Gaison, N. S. Oblath, D. Rosa De Jesús, J. R. Tedeschi, B. A. VanDevender

Pennsylvania State University

M. C. Carmona-Benitez, L. de Viveiros, R. Mueller, A. Ziegler

University of Pittsburgh

P. T. Surukuchi

University of Texas, Arlington

B. J. P. Jones

University of Washington

A. Ashtari Esfahani, C. Claessens, P. J. Doe, S. Enomoto, A. Marsteller, E. Novitski, R. G. H. Robertson, G. Rybka

Yale University

K. M. Heeger, J. A. Nikkel, P. L. Slocum, A. B. Telles, T. E. Weiss

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Executive Summary

Brent will write this after the rest of the doc is mature.

A condensed summary for the busy program manager on the go that may not be able to read the whole document (Think Linda Horton, Acting Associate Director of NP and up the org chart from there). This should stand alone without the rest of the document.

- Objective: Purpose and scope of the document. Extremely brief statement of the LFA concept and goals.
- Phase IV Goals/Mission/Vision: Need for a new m_β measurement. Need for new technology. Goals of Phase IV.
- Phase III LFA Goals/Mission/Vision: Role of this LFA in Phase IV R&D. What technology demos? What scientific outputs?

1 Introduction

Primary Author: Brent.

An overview of the document to come. Explain its purpose. Describe the layout. State the main conclusions. There should be no major surprises later in the document. This is for people who will go on to read the whole thing, in contrast to the Executive Summary.

2 Neutrino Mass

Primary Author: Walter Pettus

Why is neutrino mass interesting? A description of our current state of knowledge. Context for Project 8 in light of KATRIN and cosmology - why is a probe of IH interesting and timely? Why would this LFA potentially be interesting.

3 Project 8 Overview

Primary Author: Walter Pettus

High level description of the Project 8 approach - T and CRES. The Phase IV strawman. Phase III R&D Program.

3.1 Neutrino Mass Sensitivity

Talia, Ben

Description of KPPs that feed project 8 sensitivity and presentation of scenarios to achieve 40 meV. This should foreshadow why we will focus our R&D as described below (e.g., to reach some KPP values that we can't currently achieve)

Outline: [We may want to move item 2, and maybe item 4, to the end.]

1. Introduce neutrino mass sensitivity equation
2. Present a few (3?) Phase IV sensitivity scenarios that reach 40 meV: plots, description
3. Break down contributions to sensitivity—e.g., resolution, efficiency, and background contributions (could be based on Sensitivity presentation at Science Review, as well as the document prepared for that review)
4. On the basis of that, identify KPPs and discuss what R&D is needed to achieve Phase IV values of KPPs. Write that this R&D will be performed for, and demonstrated with, the LFA. Perhaps have a table of Phase II value vs. Phase IV value?
5. Present results of the Phase IV parameter scan, to motivate the potential paths to 40 meV.
6. Discussion of the parameter scans: Which parameter improvements are more or less challenging to achieve. (Pooling input from experts.)
7. Brief conclusion.

3.1.1 Analytic Sensitivity Formula - Talia

Project 8's sensitivity to the neutrino mass depends principally on four parameters:

1. A signal rate r_s of reconstructed electron energies;
2. A background rate r_b of false events;
3. Energy resolution contributions σ_i , where “energy resolution” is the standard deviation of a detector response function that broadens the underlying beta spectrum; and
4. Uncertainties $\delta\sigma_i$ from calibration and analysis on the resolution contributions σ_i .

The signal rate is given by $r_s = C_T V_{\text{eff}} n t$, where V_{eff} is the effective volume (source volume $\times$ total efficiency), n is the source gas number density, t is the experiment livetime, and C_T is a constant equal to the fraction of events in the last electronvolt of the spectrum (2.06×10^{-13} , for atomic tritium) divided by the tritium lifetime.

An approximate expression for Project 8's sensitivity may be derived by considering a toy beta spectrum analysis, in which we simply count beta events in a window ΔE near the tritium endpoint [1]. Fewer events in the window indicate a larger neutrino mass. This method assumes that the endpoint is precisely constrained, either from analysis of beta events in a window below ΔE , or from external measurements. The resulting neutrino mass sensitivity is

$$\begin{aligned} \sigma_{m_\beta} &= \sqrt{\left(\sigma_{m_\beta}^{\text{stat}}\right)^2 + \left(\sigma_{m_\beta}^{\text{syst}}\right)^2} \\ &= \sqrt{\frac{4}{\left(3 C_T V_{\text{eff}} n t\right)^2} \left[C_T V_{\text{eff}} n t \Delta E + \frac{r_b t}{\Delta E} \right] + 16 \sum_i \sigma_i^2(n) \cdot \delta \sigma_i^2}, \\ \text{Upper limit}(m_\beta) [90\% \text{ CL}] &= \sqrt{1.28 \sigma_{m_\beta}^2}. \end{aligned} \quad (1)$$

The conversion from σ_{m_β} to an upper limit on m_β assumes that the neutrino mass is close to zero. Note that some contributions to energy broadening depend on the density n . The optimum energy interval is

$$\Delta E = \sqrt{\frac{r_b}{C_T V_{\text{eff}} n t} + 2.35^2 \sum_i \sigma_i^2}, \quad (2)$$

where the factor 2.35 converts standard deviations to full-widths-at-half-maximum.

Equation (1) has been validated with a Bayesian analysis of Monte Carlo data, accounting for instrumental broadening of the beta-decay spectrum [2]. In addition, the neutrino mass result measured in Phase II (Section 3.2) aligns with the analytic sensitivity model [3]. We employ this model to optimize design parameters and estimate the sensitivity of future Project 8 phases.

Eq. 1 shows that the 90% CL upper limit on m_β scales approximately with (signal rate) $^{-1/4}$. Accordingly, to improve mass sensitivity by a factor of 10, 4–5 orders of magnitude increase in statistical sensitivity are needed, alongside reduced systematic uncertainties. This drives the physical volume requirements of each phase of Project 8.

Beyond placing an upper limit on m_β , Project 8 has the potential to measure m_β with a positive lower limit, thus resolving the neutrino mass scale and mass eigenvalues [1]. In Phase IV, values of m_β above ≈ 50 meV could be distinguished from zero with 90% credibility, assuming the apparatus can produce a 40 meV upper limit when $m_\beta = 0$. As an example, if the true value of m_β is 100 meV, Project 8 could resolve the mass in an interval of width ~ 40 meV (90% credibility), corresponding to a $> 5\sigma$ “discovery sensitivity” of nonzero neutrino mass. These findings are from a Bayesian analysis of Monte Carlo data [2].

3.1.2 Contributions to Main Parameters - Talia

Break down contris to signal rate (V_{eff}), background rate, resolution, and resolution uncertainty.

Signal rate: Factors of density, livetime, and effective volume. Density assumption: Contributions to electron scattering rate of gases other than atomic tritium are negligible. Effective volume—Total volume: Set by cavity frequency and number of cavities. Assume 325 MHz in calculations for Phases III and IV. Efficiency: Electron trapping efficiency, Event detection efficiency, Limits on useful radius from electrons hitting wall and from Halbach field variation.

3.1.3 Sensitivity Parameter Scans to Motivate Phase IV Scenarios - Ben

TODO some of this might go better in the Key Param for 40 meV section. The LFA design is intended to bridge the gap to from the demonstration of cavity CRES to a high precision measurement of atomic tritium with sensitivity of the neutrino mass down to 40 meV in Phase IV. In order to achieve this goal, the quantitative scaling of the sensitivity must be well understood for every contributing design parameter. These scaling metrics inform the level of improvement needed across the full design, and allow the construction of multiple scenarios for an atomic tritium experiment reaching 40 meV sensitivity. Each scenario relies on improvements in different key parameters, taking into account estimated cost and difficulty expected for such a configuration. During the development of the LFA experiment, the feasibility of achieving certain levels of improvement for each parameter will be evaluated. The most realistic Phase IV scenario with an achievable cost and timeline can then be pursued.

3.1.4 Key Parameters to Reach 40 meV Sensitivity - Talia and Ben

Phase IV scenarios go here. Values, motivation for achieving them. (May want to move this to after the param scan motivation section)

3.2 Prior Progress: Phases I and II

succinct summary of prior published results.

4 The Low Frequency Apparatus Overview

Primary Author: Elise

Comprehensive, but high-level description of LFA scope.

4.1 LFA Goals

Hamish

4.1.1 Science Goals

The LFA is primarily a large-scale technical demonstrator for use of the CRES method in measuring neutrino mass. It is nevertheless a key aspect of this role that it be constructed as a fully functional instrument capable of accepting a beam of atomic tritium when it becomes available, and carrying out neutrino mass measurements. This instrument will have sub-eV sensitivity potentially comparable to the state-of-the-art KATRIN experiment, with a source of atomic tritium. With its low background, it will be sensitive not only to active neutrino mass, but it will in addition excel in searching for light sterile neutrinos, relic neutrinos, non-standard interactions, right-handed currents, and other new physics.

4.1.2 Technical Goals

As the first large-scale CRES demonstrator there are many challenging technical goals for the LFA.

1. The magnetic design consists of 4 interlocking magnetic elements, a Ioffe coil to trap atomic tritium, a large uniform solenoidal field, trap coils for trapping electrons, and a magnetic compensation system for the earth's field and other environmental fields. These must be designed and work as an integrated system, and meet the specifications necessary for a high-resolution experiment in a large volume.
2. At low frequencies CRES signals have low powers, in the aW (10^{-18} W) range, and ultra-low-noise quantum electronics must be implemented to deliver the required signal-to-noise ratio. Not only must the 'carrier' frequency centered on the cyclotron mean frequency be detectable, but also at least one sideband for reconstructing and correcting for axial motion of electrons in the trap. The coupling of the electronics to the cavity is in itself a required demonstration. This low-frequency system must also work in an environment of electromagnetic interference from many sources.
3. A successful cryogenic design is essential. The cavity walls serve as a very large cryopump to keep the pressure of T₂, which has a higher endpoint energy than T, below the level at which it would be a background, about 10^{-4} in activity. The temperature must be $\lesssim 4$ K, requiring a (mildly) pumped He system. Periodic warming to $\simeq 20$ K to drive off the accumulated tritium for recirculation is necessary.

4. Atoms that have been cooled in an external beamline must be injected into the trap and held there for as long as possible (1000 s, roughly) to keep tritium inventory down and not to exceed the capabilities of the beamline for supplying makeup atoms. LFA demonstration will be with Li beams.
5. Comprehensive simulations underpin the LFA construction. Moreover, since readiness for tritium running but not actual tritium running is a requirement, that aspect requires full simulation, such that the transition to tritium can be confidently made.

4.2 LFA sensitivity

Talia

Constraints (before or in this section): BNL magnet fabrication, funding (limits size), noise (technology readiness).

What would be the neutrino mass sensitivity if we upgraded to an atomic tritium source. Will have presented the framework for Project 8 Sensitivity Analysis above. Present here scenario for 0.56 GHz.

Why sensitivity vs. density looks how it does. The black curve shows “Phase IV module 1,” Right: Each curve is plotted for one density—the operating density for the LFA, and the optimum density from the left plot for the other curves Why the right plot looks as it does. An SNR threshold for detection is re-optimized at each point on the sensitivity vs. density curve.

In Table 1, the 85 meV “resolution from B variation in r, ϕ, t ” is the value required to reach 40 meV sensitivity in one Phase IV scenario. For the LFA, adequate sensitivity can be achieved even with a 300 meV resolution contribution from field variation.

Table 2 lists neutrino mass sensitivities of an LFA upgraded with tritium, as well as other outputs of the sensitivity calculation. The “LFA target” values correspond to Figure 1 and Table 1. The “LFA threshold” values represent key performance parameters (KPPs) for the LFA. Threshold KPPs are chosen as follows: The threshold SNR of 80 for a 1 ms event (90° pitch angle, averaged over radii) is the approximate expected SNR in Phase IV. The 300 meV threshold resolution allows for unforeseen effects that broaden the spectrum; 300 meV is the target resolution for Project 8’s Cavity CRES Apparatus (CCA), which is presently being commissioned. This resolution worsens the sensitivity by 100 meV, compared with the target case.

In the threshold scenario, we also increase a constant multiplying the RF background rate formula by a factor of 10^3 . This constant depends on the reconstruction algorithm(s) used, and the threshold value represents a conservative view of event reconstruction. After an event detection parameter is re-optimized, the threshold background rate is $15\times$ higher than the target background, and the sensitivity worsens by 10 meV at the operating density. The threshold total efficiency results from the SNR of 80, the higher background rate constant, and a higher minimum trapped pitch angle of 88°, allowing for the possibility that the desired field broadening cannot be achieved with the target $\theta_{\min} = 87^\circ$.¹

¹Ongoing work on electron trap design will determine what minimum trapped pitch angle is acceptable from the perspective of magnetic field broadening.

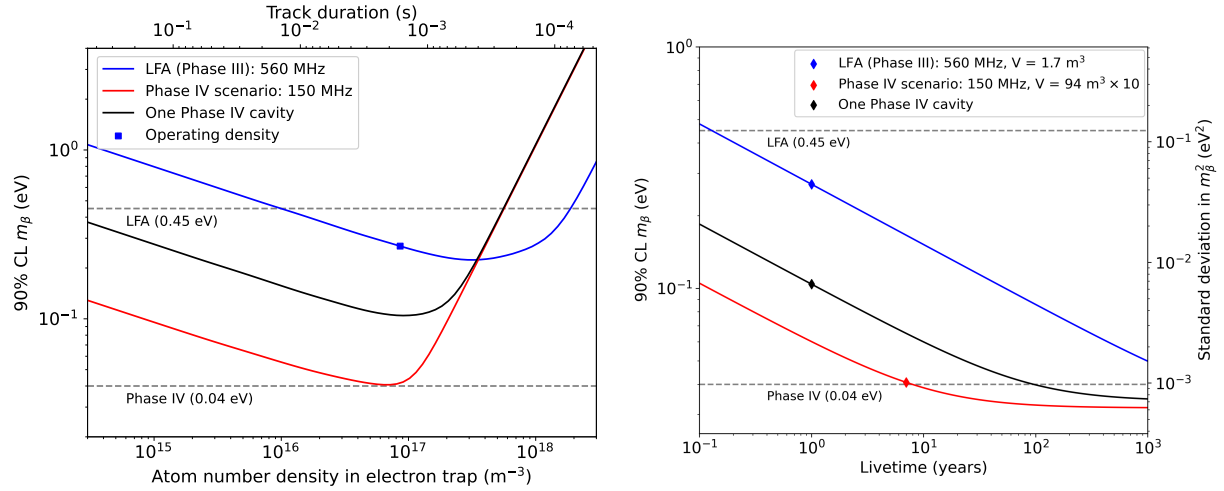


Figure 1: Left: Neutrino mass sensitivity (90% CL) vs. density of tritium atoms in the electron trapping region, if the LFA were upgraded to include a tritium source. The legend indicates cavity resonant frequencies. The LFA operating density is limited by achievable cryopumping speeds to a value below the optimum. Right: Sensitivity vs. livetime curves, each plotted for one density. The legend indicates physical volumes of cavities. LFA curves correspond to Table 1 and the “LFA target” column in Table 2. The dashed line at 0.45 eV represents the approximate LFA sensitivity if all threshold Key Performance Parameters are achieved; see Table 2. The red (black) curve has a seven (one) year livetime.

Questions: Should I break this into two figures, so that each is larger? Should I switch out the right plot for one with sensitivity vs. $N_{\text{cavities}} \times \text{livetime}$? Please see slides on BC from 1/10/25.

Possible effect of neglected contributions to the energy resolution?

Add a paragraph summarizing risks and uncertainties—i.e., how hard it is to attain the scenario described here.

Table 1: Input values for calculating the neutrino mass sensitivity of the LFA, if the experiment were upgraded to include an atomic tritium source. Names of key performance parameters (KPPs) are bolded. Most KPPs are given in Table 2.

Input Parameter	LFA Value	Relevant Section(s)
Livetime	1 year	6
Background field strength	20.75 mT	5.8
Cavity length/diameter	7.65	5.2
Minimum trapped pitch angle	87°	5.10
Electron trap length	4.05 m	5.10
Flat-fraction of axial trapping field profile	0.75	5.10
Resolution from B variation in r, ϕ, t	85 meV	5.5, 5.7, 5.9, 5.10, 5.12, 8.2
Source gas temperature in cavity	4 mK	5.11, 5.13, 5.14, 5.15
Cavity temperature	4 K	5.2, 5.3, 5.15
Cavity unloaded Q	10^5	5.2
Amplifier temperature	20 mK	5.3
Amplifier efficiency	0.5	5.3
Line attenuation	−0.1 dB	5.3
Circulator and microwave pump attenuation	−1.3 dB	5.3
Average power fraction: (carrier)/(90° e [−])	0.40	5.2, 5.10
Sideband order detected	2	8.1.2, 8.2
Average power fraction: (sideband)/(90° e [−])	0.10	5.2, 5.10
Radial Ioffe bite—unusable dist. from walls	0.041 m	5.11
Rate of cosmic ray–T interactions per atom	2.4×10^{-29} /eV/s	5.11
Uncertainty on energy resolution	2%	5.5, 5.12, 8.2, 8.3

Table 2: Outputs from a calculation of the LFA’s neutrino mass sensitivity, if it were upgraded to include an atomic tritium source. The “LFA target” column shows results calculated from the inputs in Table 1. These are “reach” parameter values, calculated for the listed operating density (third row). The “LFA threshold” column contains values of KPPs that should be achieved in the LFA. Names of KPPs are bolded.

Output Parameter	LFA Target	LFA Threshold	Relevant Section(s)
Limit on m_β, 90% CL	0.27 eV	0.44 eV	
Optimum tritium density in electron trap	$3.8 \times 10^{17} / \text{m}^3$		5.3.3, 8.2
Operating density, to reduce atom losses	$8.6 \times 10^{16} / \text{m}^3$		5.11, 5.14, 6
Number of T atoms in electron trap	1.2×10^{17}	1.2×10^{17}	5.11, 5.14, 6
Total background rate	$2.2 \times 10^{-10} / \text{eV/s}$	$3.2 \times 10^{-9} / \text{eV/s}$	5.3, 8.1.2
Cavity frequency	560 MHz		5.2
Cavity loaded Q	490		5.2
SNR of 1 ms event ($\theta = 90^\circ$)	250	80	5.2, 5.3
SNR detection threshold for track	22		5.3.3, 8.1.2
Trap q , in $f_c = f_c^{B_{\min}}[1 + q(\pi/2 - \theta)^2]$	0.17 /rad ²		5.10
Trap p , in $f_a = f_c^{B_{\min}}p[\pi/2 - \theta]$	0.019 /rad		5.10
Total energy resolution, stdev.	86 meV	300 meV	8.3
From $\sigma_f^{\text{carrier}}$ and $\sigma_f^{\text{sideband}}$ due to noise	11 meV		5.3, 5.10
From field variation and reconstruction	85 meV		5.9, 5.10, 5.12, 8.2
From thermal Doppler effect	1.5 meV		5.11, 5.13, 5.14, 5.15
Total efficiency	0.029	0.011	8.3
Trapping efficiency	0.049		5.10
Detection efficiency	0.76		5.3.3, 8.1.2
Efficiency from limits on useful radii	0.76		5.11

5 LFA Conceptual Design

Primary Author/Coordinator: Elise w/ Working Group Chairs

There will be subsections (and subsubsections) for each major system (subsystem). For each major subsystem of the LFA, use this template: 5.1 and address the following topics: Concept of operations. Analysis of alternatives. TRL designation for each.

The major subsystems correspond to Working Groups:

- CRES Cavity **Juliana**
- Magnetic Field (possibly split into separate subsections) **René**
 - brief intro
 - environment shielding
 - Background field + stabilization
 - shimming
 - CRES trap
 - Ioffe trap + endcap
 - field mapping
- CRES Calibrations
- Lithium beamline
- Cold Electronics **Wouter**
- Warm Electronics (likely to merge this straightforward section with Cold Electronics) **Noah**
- Gas Handling
- Cryogenics and Infrastructure

5.1 LFA Subsystem Template

The first part, at this subsection level, should be a brief introduction that summarizes what is to be given in more detail in the subsubsections below.

5.1.1 Subsystem Objectives

Describe the function that this subsystem serves in the LFA. For example, “The Lithium Beamline delivers lithium atoms into the cavity ... is critical for demonstrating simultaneous atom trapping and CRES measurements.” If known state the conditions (key performance parameters or KPPs) that must be satisfied. E.g., “the lithium beamline must deliver X atoms per second at a temperature of Y ...” or “field uniformity of σ is required over a volume V to achieve the sensitivity discussed above.”

5.1.2 Operational Principles

A succinct presentation of whatever background physics is required to understand how this system works. E.g., If you were writing about the CRES cavity, you would describe how the electron excites the cavity and how the information we want is encoded in that excitation (easy for me say, right?)

In some cases, there may only be one way to do things. Or, there are many ways to do things but we have already chosen one. Describe the chosen method here. In cases where we are still evaluating alternatives, state how we will choose in the end and then describe those alternatives below. If we've already chosen among multiple options, describe the alternatives we did not choose and explain why they are inferior. E.g., if we've definitely chose Ioffe traps, then describe that here, and then say why we didn't choose Halbach arrays below.

5.1.3 Analysis of Alternatives

See above. These are descriptions of all the options from which we need to choose one, or a description of all the options we did not choose. We don't have justify crazy things that were never really a possibility, but our longtime observers and reviewers will want to know how we made choices. E.g., whichever way we go with Ioffe versus Halbach, about half of the community will think (because they heard us say it at some point) that we were going to do the other.

5.2 CRES Cavity

5.2.1 Subsystem Objectives

The cavity is the holding vessel for the tritium atoms as well as the detector for the cyclotron radiation emitted by the decay electrons, the heart of Project 8's "source = detector" concept. The LFA will initially be operated using a cold beam of lithium atoms, but needs to be capable of an upgrade to an atomic tritium beam once it becomes available. Thus, the cavity needs to be fully compatible with both lithium and tritium. Further, due to atomic trap losses in either configuration, there will be buildup of (a) lithium, and (b) tritium and helium at the walls. The cavity thus needs to be able to undergo temperature cycling to remove the buildup. To be compatible with magneto-gravitational atom trapping, the cavity needs to be (a) tall, such that the cold atoms do not leave at the top; and (b) thin-walled to allow for the magnetic field from the multipole magnet surrounding the cavity to be sufficiently strong inside of the cavity wall. Further, as the atom-trapping field falls off quickly towards the inside of the cavity, some region close to the wall will have a non-uniform magnetic field, and thus be not useful for CRES. This calls for the cavity to have a large enough diameter, such that this outer, non-uniform-field region is small compared to the total cavity volume. However, the cavity walls also need to be sufficiently thick to be stable against pressure gradients, which might build up if a sub-component fails. The cavity must then be stable enough to prevent a tritium spill. Under normal operating conditions, both the cavity and the volume immediately surrounding it will be evacuated, with the cold atomic beam being injected into the cavity.

The LFA is the first Project 8 endeavor in large, cubic-meter CRES volumes. Previously demonstrated CRES readout from waveguides is not feasible at that scale. Further, the atomic trapping constraints have identified lower CRES trapping fields B_{CRES} to be strongly preferred in order to keep trapping lifetimes sufficiently high. Cavities provide a naturally large volumes at low resonance frequencies. The power emitted by trapped electrons via CRES scales as $P \propto B_{\text{CRES}}^2$, leading to challengingly low expected collected powers. With their resonant environment, cavities enhance the power emitted by electrons if their frequency coincides with the resonant frequency of a mode. This makes cavities uniquely suited for large-volume CRES experiments.

The precise determination of the CRES frequency requires a sufficiently long observation time. Thus, the CRES field needs to be extremely stable. While this is mainly a requirement for the magnetic subsystems, it affects the cavity design as well. To enable the needed observation time, the electron is trapped magnetically. This trapping field needs to be flat over a large region, which is easier accomplished with long and skinny traps, thus also requiring the cavity to have a long and skinny shape. Further, the cavity and readout must not distort the magnetic field experienced by electrons.

To achieve a simple signal structure, we require the RF photons emitted by the electrons to only drive one resonant cavity mode. All trapped electrons should be visible in a defined energy range close to the tritium endpoint. This requires the mode bandwidth to be well matched to the frequency spread of the emitted radiation that depends on the electron's energy and experienced magnetic field.

The cavity is an open system: at the bottom, it is connected to the atomic beam line, supplying fresh, cold, trappable atoms into the cavity volume. This bottom is cone-shaped due to the magnetic field requirements for the atoms in that region. At the top, it is connected to turbo-molecular pumps to recover non trappable atoms and molecules. The top also contains openings for calibration devices, such as electron injection sources and laser guides. Along its perimeter, the cavity is surrounded by the multipole atom trapping magnet. The signal is read out in situ in the cavity, either via a connection at the top of the cavity, where pickup structures do not obstruct atoms, or from the center where the cavity mode is strongest. This signal is then transported out of the cavity via the open top and connects to cryogenic amplifiers. As the entire cavity needs to be held at cryogenic temperatures, it is also embedded in a large cryogenic system.

5.2.2 Operational Principles

In free space, an electron in a magnetic field radiates the power [4]

$$P(\theta) = \frac{2\pi e^2}{3\epsilon_0 c} f_c^2 (\gamma^2 - 1) \sin^2 \theta, \quad (3)$$

with the electron's pitch angle θ , Lorentz factor γ and cyclotron frequency f_c . The latter is given by

$$f_c = \frac{1}{2\pi} \frac{eB_{\text{CRES}}}{\gamma m_e}. \quad (4)$$

Thus, the radiated power is $P \propto f_c^2 \propto B_{\text{CRES}}^2$ where B_{CRES} is the external magnetic field inducing the cyclotron motion. From constraints related to atomic tritium handling, B_{CRES} should stay considerably below the 1 T at which CRES has thus far been demonstrated [5,6], leading to substantially lower radiated powers: at $B_{\text{CRES}} = 0.04$ T, $f_c = 1$ GHz, and $P = 1$ aW (10^{-18} W).

Those low radiated powers pose a challenge to detect electrons and reconstruct their energies, however, the power can be enhanced in the resonant environment by the Purcell effect [7] if the electron cyclotron frequency is close to the frequency of a cavity mode.

In a cylindrical cavity, the resonant modes are either transverse electric (TE) or transverse magnetic (TM), with the electron coupling well to the TE modes. The TE_{011} mode maximizes the mode volume while also having a particularly simple signal structure. It's frequency is given by

$$f_{011} = \frac{c}{2\pi} \sqrt{\left(\frac{X'_{01}}{R}\right)^2 + \left(\frac{\pi}{L}\right)^2}, \quad (5)$$

where X'_{01} is the first root of the derivative of the zeroth Bessel function, and R and L are the cavity radius and length, respectively. Signals from electrons radiating at $f_c = f_{011}$ will be enhanced, while the radiation from off-resonance electrons will not excite the cavity mode. Thus, signals from electrons deep in the spectrum, which do not carry information on the neutrino mass, will be suppressed.

The magnetic trapping of the electrons, necessary for an accurate frequency determination, changes the average magnetic field $\langle B_{\text{CRES}} \rangle$ experienced by the electrons depending on their pitch angles θ and the geometry of the trapping field. An electron with a pitch angle of $\theta = 90^\circ$ will always stay at the bottom of the trap and experience no modification, while an electron close to the minimal trappable pitch angle, θ_{min} , will climb up the walls of the trapping field before turning around and experience $\langle B \rangle > B_{\text{CRES}}$. This axial motion will lead to two observable modifications of the signal:

1. Sidebands appear for all but 90° -pitch angle electrons, and the modulated signal now has frequency components $f_c, f_c \pm f_a, f_c \pm 2f_a, \dots$ where f_a is the axial frequency.
2. The carrier frequency f_c increases as $\langle B \rangle > B_{\text{CRES}}$.
3. The odd sidebands are suppressed for an axially centrally located receiver.

To reconstruct the electron's energy, at least the carrier frequency and one sideband needs to be detected. As the carrier frequency increases with increasing pitch angles, as does the difference between carrier and sideband, it is advisable to focus the detection on the carrier and the lower sideband, as that will lower the minimal mode bandwidth needed. The electron signal and cavity detection scheme are sketched in Figure 2. The TE_{011} mode is strongest in a cone approximately at half the cavity radius and falls off to the center and the cavity walls. Axially, the mode is strongest at the midpoint falling off towards the top and bottom of the cavity which terminate the mode. The radial field strength variation makes the reconstruction of the radial position of the electron possible. This in turn yields a more precise energy reconstruction as the magnetic field also varies radially which needs to be

corrected for. Due to the extent of their cyclotron orbits, even electrons orbiting around the center of the cavity will excite the mode.

The simplest readout involves one pickup antenna located at the midpoint. However, the signal readout must not interfere with the coils of the atom trapping magnet that are snugly fit around the cavity. Therefore, any readout design will have to exit the cavity at the top. The top of the cavity will also be the transition between the tritium volume and the tritium-free volume. The optimization of the readout design is currently one of the priority R&D topics related to cavity design.

- some discussion of frequency choice (and the multiple cavity detectors concept) with reference to magnetic fields section
- frequency scaling, prototypes and demonstrators, and how this affects TRLs.

5.2.3 Analysis of Alternatives

Previously, an antenna array concept was explored for Project 8, leading to a mature design and prototype multiplexed antenna array ring [Cite antenna array paper once on arxiv](#). Cavities offer a much simpler signal readout design, requiring one to a few channels as compared to thousands for the antenna array, and were ultimately chosen as the more viable path for Project 8.

We have so far focused our discussion on the TE_{011} mode, with an extremely simple signal structure necessitating only one readout probe. We may realize that the resulting signal is too simple for a full reconstruction of the electrons' properties. In particular, radial reconstruction using only the radially varying mode amplitude (and therefore signal power and frequency chirp) may be insufficient. This might lead us to explore modes that allow for better radial reconstruction, either in a single-mode cavity or where the electron can excite multiple modes.

Another interesting design is a stack of etalons. Together, they would have the same volume of a cavity with similar frequency of the TE_{011} mode, but would allow for independent mode volumes and readouts. For long and thin cavities, the mode separation between subsequent TE_{01p} modes gets small, limiting the possible bandwidth of the TE_{011} mode. Thus, the length of the cavity and with it the gas volume cannot be arbitrarily large. A stack of etalons solves this problem. Further, each etalon has a fraction of the volume, leading to a smaller mode volume, and in turn a higher-power signal. While not the current baseline design, the etalon concept will be explored within the CCA testbed.

In the CCA, initially the readout is a single probe located at center of the cavity axially, where the TE_{011} mode is maximal. As noted above, this readout is blind to first order sidebands. From simulations with CST, we expect that shifting the electron trap wrt. the readout the first order sidebands will be enhanced. This will be tested in the CCA setup in order to help determine the optimum readout placement for the LFA.

5.3 Cold Electronics **Primary Author: Wouter**

The objective of the LFA system's cold electronics is to read out the cavity signal with a minimal amount of added noise to enhance detection efficiency. First, we will describe the noise sources, and then we will discuss the CRES signal characteristics. Both of these give event-by-event signal-to-noise (SNR) information that sets the detection efficiency, false event rates, and effective volume. In the following, two scenarios are compared, one with a resonance at 500 MHz and $L/D \approx 5$, and a design with a resonance at 1 GHz and $L/D \approx 9$.

Add the default scenario with 560 MHz, $L/D \approx 7.6$, $T_{cavity} \approx 4$ K and $T_{cold} \approx 10$ mK.

The design of the LFA cold electronics is guided by Mermethid sensitivity calculations, which include signal and noise models, and RF cascading models constructed using scikit-rf.

5.3.1 Noise temperatures

Reasonable input choices The cryogenic HEMT amplifier in both scenarios is the LNF-LNC0.2_3B with noise temperature (1.4 ± 0.1) K for both frequencies. We assume a quantum efficiency of 50%, corresponding to 2 added noise photons for quantum amplifiers. This is a conservative estimate but does not limit the total noise budget in most scenarios.

The internal quality factor – Q_{int} – is taken to be 1×10^5 . Potential R&D into superconducting cavities can lift us above 1×10^6 , into the region where further improvements in Q_{int} do not impact the total readout noise.

The circulator we use to probe the S_{11} response and isolate the cavity impedance mismatch from the amplifier can be produced by Quinstar. The insertion loss at 500 MHz is ≈ 0.8 dB (QCY-M4505001xx) and ≈ 0.5 dB at 1 GHz (QCY-G0080121xx).

The cables connecting the cavity readout and the circulator and connecting the readout parts further upstream till the HEMT will be superconducting. An example part is Keycom's UPJ07 NbTi-NbTi superconducting coaxial cable. The cable itself has an insertion loss of < 0.005 dB m $^{-1}$ which is negligible, however, the connection between the cavity readout and the coax will have some insertion loss. We currently do not have a reasonable estimate and will take this as 0.1 dB.

When we use quantum amplifiers, there are two generic mechanisms to drive them: flux pumping and microwave pumping. For the Travelling Wave Parametric Amplifiers (TWPAs) investigated at MIT, a microwave pump needs to be inserted. Coupling this tone into the amplifier requires a directional coupler that adds additional insertion loss. However, because this insertion loss is at the coldest stage, it has a small impact on the noise budget. For this exercise, we assume an additional 0.5 dB of loss. When operating only with a HEMT, no directional coupler is needed.

Thermal decoupling / radiative cooling of the mode photon population In the default LFA scenario, there is a dilution fridge – we assume a base temperature of 15 mK – acting as a cold reservoir to cool the photon population in an over-coupled readout. We benefit from a strongly over-coupled cavity, as set by $g = Q_{int}/Q_{ext}$. Note that the loaded quality factor

sets the width of the resonance:

$$\frac{1}{Q_{loaded}} = \frac{1}{Q_{int}} + \frac{1}{Q_{ext}}, \quad (6)$$

and impacts total signal power through resonant enhancement, see section 5.3.2. For the 1 GHz design, $g \approx 65$ while for the 500 MHz $g \approx 100$. This difference is driven by the L/D requirement that impacts the necessary width of the resonance.

This thermal decoupling effect is most important when the circulator is at a physically different temperature than the cavity. In the alternative scenario where we have no dilution fridge and use a HEMT amplifier, there is no cold reservoir to cool the photon population in the mode. However, the noise emitted by the back of the HEMT might still lead to a small but noticeable cooling effect corresponding to a cold bath of ≈ 2 K.

Contributions to the system noise temperature When designing the LFA cryogenic readout electronics, it is important to understand the different contributions to the noise:

1. Black body photon population inside the cavity at physical temperature T_{cavity} .
2. Coupling between the cavity and the transmission line.
3. Attenuation over the transmission line.
4. Noise generated in the circulator/isolator reflecting off the cavity.
5. Noise added by the amplifier at physical temperature $T_{amplifier}$

If we use superconducting cables, the loss over the transmission line can be ignored and the noise power for a small $\Delta\nu \ll f_{cyclotron}/Q_{loaded}$ near resonance is:

$$\begin{aligned} P_{noise}(\Delta\omega) = & k_B T_{cavity} \Delta\nu \frac{4g}{(1+g)^2} \frac{1}{1 + \left(2Q_{loaded} \frac{\Delta\omega}{\omega_0}\right)^2} && \text{(cavity generated noise)} \\ & + k_B T_{isolator} \Delta\nu \left(1 - \frac{4g}{(1+g)^2} \frac{1}{\left(1 + 2Q_{loaded} \frac{\Delta\omega}{\omega_0}\right)^2} \right) && \text{(reflected isolator noise)} \\ & + k_B T_{amplifier} \Delta\nu && \text{(amplifier noise)} \end{aligned}$$

Here, $\Delta\omega$ is the frequency, near resonance, at which the noise is evaluated.

The noise power can be converted to an effective system noise temperature using $P_{noise} = k_B T_{system} \Delta\nu$.

At the moment, the framework only considers signal emitted on-resonance but this should be sufficient to compare different scenarios. Extensions to include the effect of mode-structure and power in sidebands is in progress.

Table 3: Summarising the noise budget for different configurations in the case of a non-superconducting cavity. See text for assumptions. Cost per channel excludes the dilution refrigerator. Q refers to near-quantum-limited amplifier

Contribution	1 GHz, HEMT	500 MHz, HEMT	1 GHz, Q	500 MHz, Q
4 K Cavity	4 K	4 K	0.23 K	0.15 K
Cavity-coax loss	0 K	0 K	0.14 K	0.15 K
Circulator	0 K	0 K	0.01 K	0.01 K
First stage amp	1.4 K	1.4 K	0.10 K	0.05 K
System noise	5.4 K	5.4 K	0.48 K	0.36 K
Cost per channel	\$10k	\$10k	\$30k	\$30k

5.3.2 Approximate signal power and features

The signal emitted by an electron into the TE_{011} mode is analytically understood and enhanced by the loaded quality factor. However, the loaded quality factor needs to be sufficiently low to enable over-coupling (see section 5.3.1 and cover the required bandwidth set by:

- The variation in carrier frequency for different pitch angles and radii due to the mean magnetic field variation.
- Comprise the first and second-order lower sidebands, at the minimum on one side.
- Enable reconstruction of a sufficiently large range of electron energies below the end-point region, here taken to be 100 eV, leading to a sub-dominant effect.

A heuristic flat-harmonic analytical trap shape is used to estimate the maximal loaded quality factor with a flat-fraction shrinking for lower L/D values. Combined with a 3° pitch angle trapping range, $Q_l \approx 750$ for the 1 GHz scenario and $Q_l \approx 500$ for the 500 MHz scenario. Ultimately, these numbers need to be tuned using a more advanced model of the magnetic trap and sideband-power distribution from simulations.

Radiated power in TE_{011} mode Figure 3 shows the radiated power comparisons for both scenarios, assuming that all power is emitted on resonance in the TE_{011} mode and employing a simplified analytical trap model.

The outcomes conform to the expectation that the cyclotron power – as described in Hanneke’s thesis and ignoring the axial motion effects – scales as $P_{CRES} \propto Q_{loaded}/V$. For the two scenarios, we find:

$$\frac{P_{1\text{ GHz}}}{P_{500\text{ MHz}}} = \frac{Q_{loaded,1\text{ GHz}}}{Q_{loaded,500\text{ MHz}}} \frac{(L/D)_{500\text{ MHz}}}{(L/D)_{1\text{ GHz}}} \left(\frac{f_{cyclotron,1\text{ GHz}}}{f_{cyclotron,500\text{ MHz}}} \right)^2 = \frac{750}{500} \frac{5}{9} 2^2 = \frac{10}{3}. \quad (7)$$

Sideband structure Up to here, the ongoing description ignored the fact that the CRES power is split between sidebands and that the signal is suppressed when emitted off-resonance. The latter effect can be modeled using:

$$P(\theta_{pitch}, Q_{loaded}, \Delta\omega) \propto \frac{Q_{loaded}}{1 + \left(2Q_{loaded}\frac{\Delta\omega}{\omega_{mode}}\right)^2} f(\theta_{pitch}) \quad \text{with} \quad \Delta\omega = \omega_{rad} - \omega_{mode} \quad (8)$$

It remains to be seen using CST if the sideband power is suppressed according to its own frequency or the frequency of the carrier. Assuming a hypothetical power distribution of 60% in the carrier, 5% in the first order lower sideband, and 15% in the second lower sideband, fig. 4 shows how the radiated power is modified depending on the relative frequency to the mode center.

Using this formalism, we can optimize the loaded quality factor and the offset between the carrier of an endpoint electron with $\theta_{pitch} = 90^\circ$ and the center of the TE₀₁₁ mode. This optimization is shown in figs. 5 and 6. Integrating this optimization with the SNR, detection efficiency and sensitivity estimate in Mermithid is WIP.

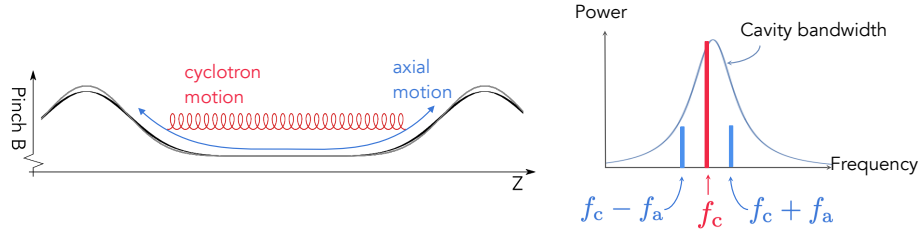
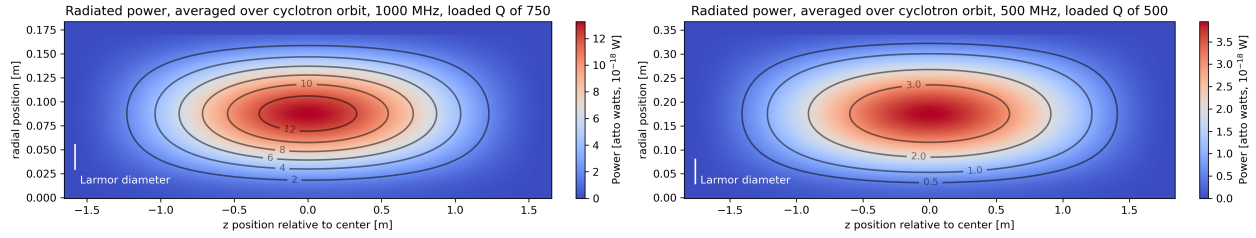
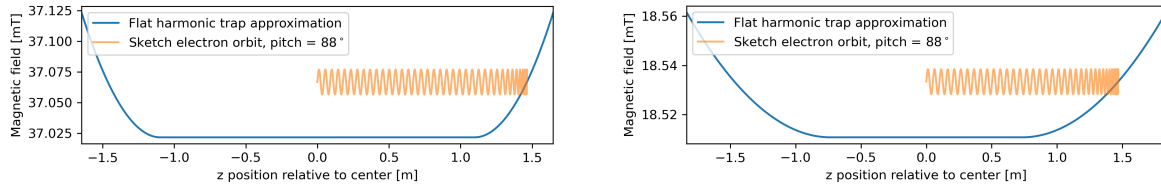


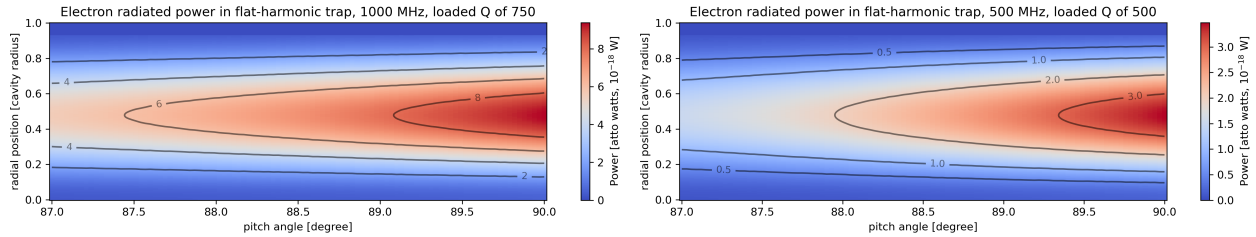
Figure 2: *Left:* The electron motion in a magnetic trap consists of the cyclotron motion (red) with frequency f_c perpendicular to the magnetic field, and the axial motion (blue) with frequency f_a parallel to the magnetic field. *Right:* The axial motion leads to the appearance of sidebands at $f_c \pm n \cdot f_a$. As the axial motion increases, the sidebands move away from the carrier, and the carrier frequency increases. Sideband detection is necessary for the estimation of the electron's experienced magnetic field, and therefore a precise energy reconstruction. The bandwidth of the cavity has to be chosen appropriately to detect carrier and a sideband.



(a) CRES power radiated by electron, ignoring axial motion and sideband structure. Using Hanneke's thesis, assuming all the power is emitted at the center of the TE_{011} mode.



(b) Heuristic trap shape assumption for $L/D=9$ (left) and $L/D=5$ (right). The analytical solution of the axial motion allows to average the power over the electron trajectory.



(c) Radiated power – averaged over axial motion – as a function of electron pitch angle at the bottom of the trap and radial position of the cyclotron center.

Figure 3: CRES signal power estimation using a simplified trap shape and no sideband structure. Radiation is assumed to be emitted in resonance with the cavity mode for all electron parameters (to be improved). On the left, plots are given for the 1 GHz scenario with $L/D=9$, and on the right, the 500 MHz scenario with $L/D=5$ is given. Need to be divided by 2 following Rick's calculations.

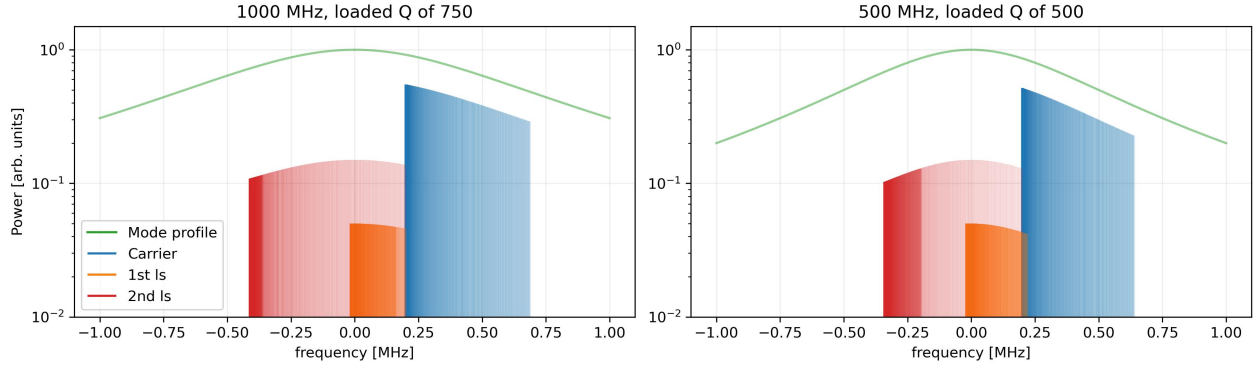


Figure 4: Location and relative power of the carrier and first two lower order sidebands. Higher frequency sidebands are omitted. Color depth corresponds to the event density, assuming a uniform pitch angle distribution. Absolute power effects arising from axial motion in the mode are yet to be included here.

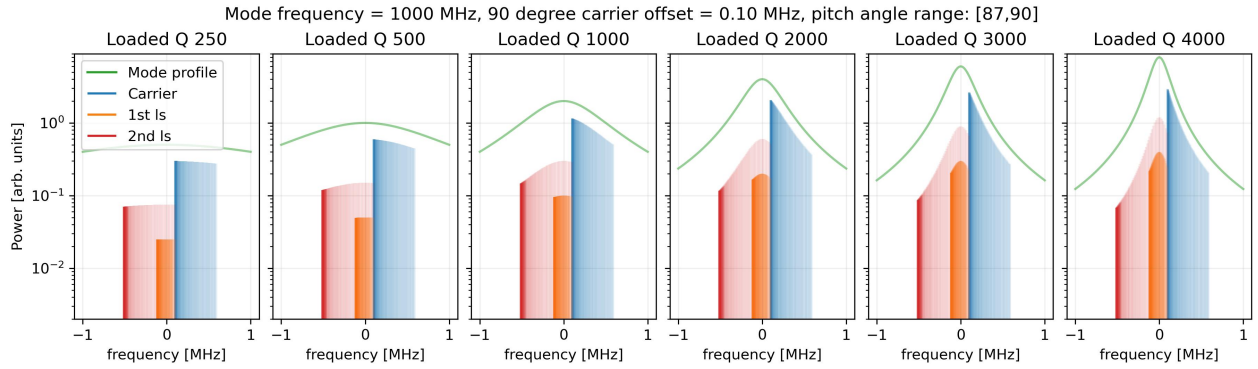


Figure 5: Effect of the loaded quality factor on the relative power of the CRES signal. An offset of 0.1 MHz was chosen to enhance the weakest (first-order) sideband maximally. Illustrated for the 1 GHz scenario.

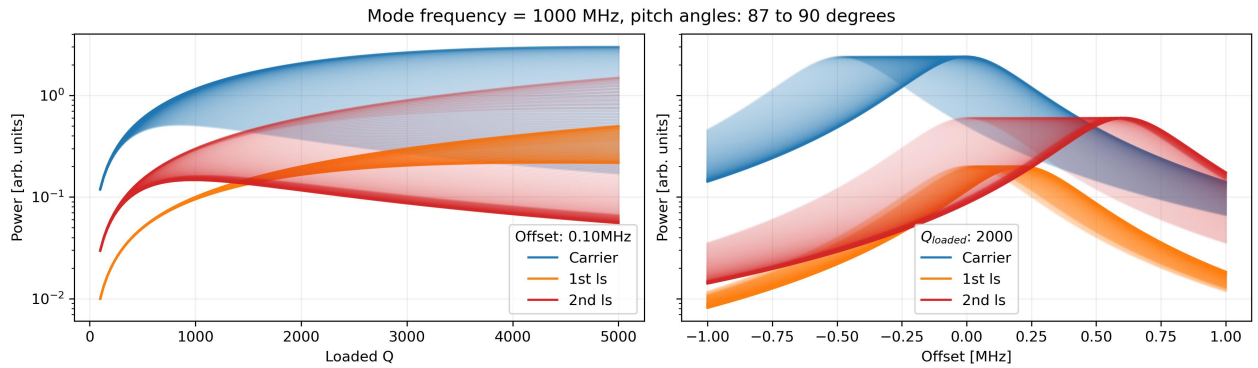


Figure 6: Scan to identify the optimal loaded quality factor (left) and the optimal offset between the mode center and the CRES radiation (right). Illustrated for the 1 GHz scenario.

5.3.3 SNR impacts on detection efficiency and effective volume

To optimize the design of the LFA parameters, we need to understand the detection efficiency and false event rate for different electron parameters. Several steps are roughly understood but have yet to be implemented.

- include absolute power in the off-resonance effects
- include a realistic pitch angle distribution
- optimize loaded Q using the resonant enhancement instead of just picking the sum of the axial and average effects.
- Decide on a good bin width (1/ scattering time? 1 eV?) to present SNR values.
- Use Rene's framework to obtain an event-by-event detection efficiency as a function of $\text{SNR}(\theta_{pitch}, \text{radius}, E_{kin})$.

5.3.4 Status of quantum amplifiers for the LFA

At MIT, quantum amplifiers are being developed at various frequencies using different technologies. The default option for the LFA is a traveling wave parametric amplifier produced using an aluminum process using the MIT.nano facilities. A preliminary design is shown in fig. 7.

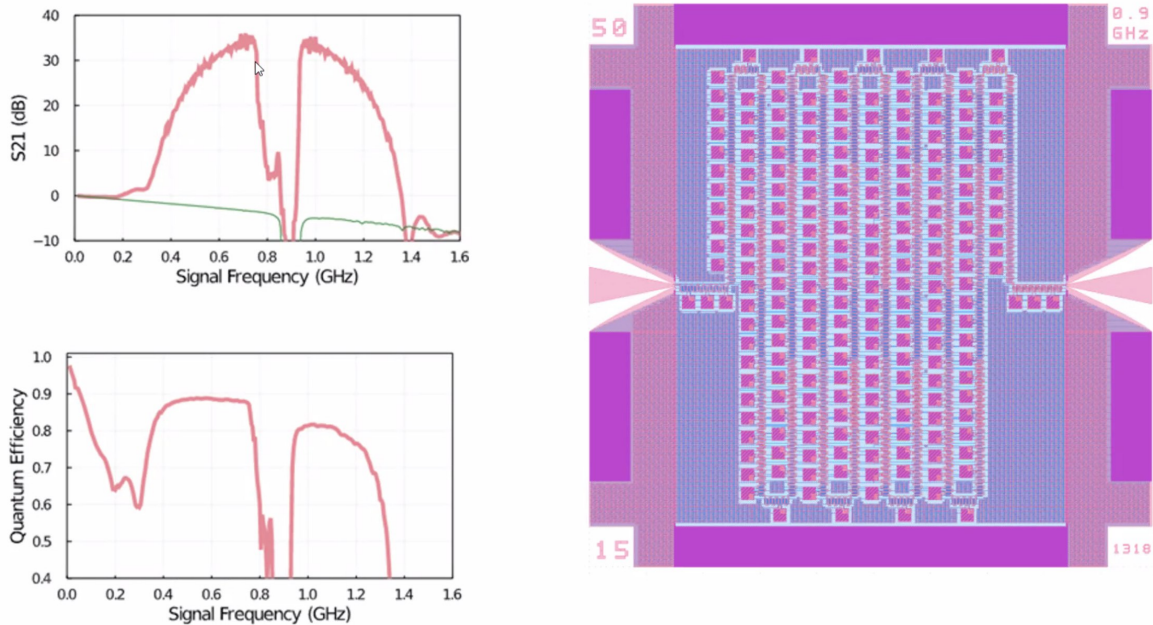


Figure 7: Simulation and design of a low-frequency TWPA by MIT's Quantum Coherent Electronics Group.

5.3.5 Costs and infrastructure estimates for dilution refrigerator integration

We need a cold bath if we want to lower the total system noise temperature significantly below the 4 K temperature of the cavity. Bluefors offers systems that can go to 900 mK, 500 mK and 10 mK. The price increase for the 10 mK system is minimal, which makes it the straightforward choice. The full current cost without educational discounts and any upgrades for the **Bluefors LD250 dilution refrigerator system** is **\$480k**.

Such a dilution refrigerator has a stainless steel can at 300 K and copper cans at 50 K, 4 K and 1 K to prevent radiative heating towards the cold stages. For the quantum scenario, we need to connect the 4 K cavity with the top or bottom of the 4 K copper plate inside the dilution refrigerator. Additional infrastructural costs will be coming from the design and construction of a coaxial double-walled (300 K and 50 K) vacuum transfer region for the superconducting RF cables. Thermal calculations must guide if a third isolation transfer tube at 4 K is needed. A mK multi-meter connection between two dilution refrigerators has been demonstrated at ETH Zürich – see fig. 8 – and should not be too difficult but will add planning and cost.

5.3.6 Analysis of Alternatives

A full reconstruction study will clarify the SNR requirements to reach the LFA sensitivity goals. As the signal can only be tuned to a limited extent, the majority of improvements compared to the CCA need to come from a drastically reduced noise temperature. The ideal scenario combines a near-quantum-limited amplifier with a cold bath of ≈ 10 mK.

If the fabrication of a TWPA at the desired frequency is unsuccessful, then we can pursue alternative superconducting amplifiers such as Josephson Parametric Amplifiers (JPAs) and Microstrip SQUID Amplifiers (MSAs). If we move to even lower frequencies for the Phase III pilot experiment, an additional technology, the RF Quantum Upconverter (RQU) would be another alternative, potentially opening up the option of multi-channel multiplexing into the microwave C-band.

Additionally, if the dilution fridge integration would somehow be infeasible, then we can operate with niobium-fabricated quantum amplifiers. However, it remains an open question if any amplifier with a physical temperature above ≈ 1 K can provide sufficient mode cooling to reach SNR values required for event-by-event reconstruction with the targetted energy resolution.

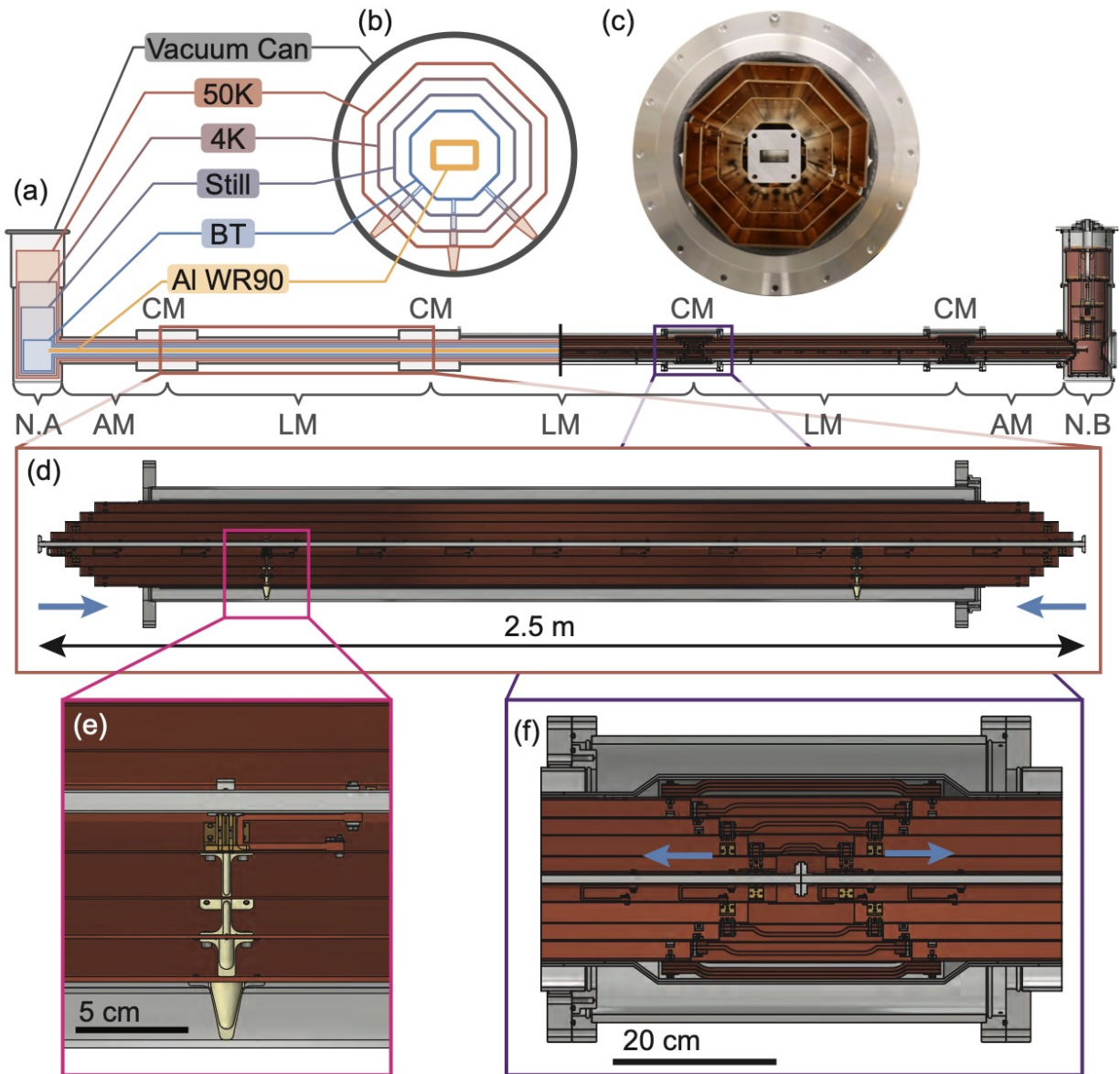


Figure 8: (a) Longitudinal cross-section of a schematic representation (left half) and a 3D model (right half) of a cryogenic link system. (b) Schematic representation, and (c) picture of a transverse cross-section of a link element. (d) Longitudinal cross-section of the 3D model of a link module. (e) Enlarged cross-sectional view of the set of posts which mechanically support the shields of the link module. (f) Enlarged cross-sectional view of the set of a connection between two link modules. The blue arrows in (d) and (f) show the direction in which the copper shield ideally move during a cooldown due to thermal contraction. N.A/N: node A/B; AM: adapter module; LM: link module; CM: connection module. Credit: Paul Magnard / ETH Zürich

5.4 Data Acquisition (Warm Electronics)

Primary Author: Noah

- Scope overview
- Goals
- Design principles
- Possible alternatives

5.4.1 Subsystem Objectives

The purpose of the Warm Electronics subsystem is to take the signal from the Cold Electronics, process the data without introducing significant additional noise or distortion, and write the data to disk. The subsystem includes the analog and digital processing of signals that emerge from the experiment, and the recording of those data to disk.

5.4.2 Operational Principles

The analog signal processing isolates and amplifies the signal in the frequency band of interest, and prepares it for digitization. The signal is amplified so that it takes advantage of the digitizer's full dynamic range – too much amplification will result in compression of the signal at large amplitudes, while too little amplification can reduce the overall SNR. The signal is also filtered to isolate the region of interest for the CRES signals in the frequency domain. This avoid the introduction of out-of-band noise, especially noise due to aliasing from frequencies above the digitizer's Nyquist limit. Typically the analog signal processing is implemented as a series of one or more filters and amplifiers carefully chosen to preserve the signal quality and match impedances between components.

The digitizer's properties determine many of the choices made for both analog and digital signal processing. The dynamic range of the digitizer is the range of voltage amplitudes that can be measured. It should be matched to the signal amplitude to give the best resolution in measuring the signal. The number of bits with which the signal is digitized, combined with distortions added to the signal by the inherently non-ideal digitization process, determine the resolution with which voltages are measured. The SNR obtained with a full-scale signal and the noise introduced by the digitizer (distortions + digitization noise) determine the effective number of bits (ENOB) for the digitizer. The digitizer should be chosen such that the added noise is small compared to the noise already present in the experiment signals. Finally, the frequency bandwidth of the digitizer is from DC (or near DC) to the Nyquist frequency, where the latter is half of the digitization rate. The experiment signal frequency band should fit comfortably within the digitizer bandwidth (unless higher Nyquist zones are being used, but that will not be the case for the LFA).

Finally, digital signal processing (DSP) is performed before the data are written to disk. While most of the fundamental signal and noise properties of the data are fixed at this point, the DSP stage can include a wide range of signal processing options. For the LFA,

this will include down-sampling and filtering the data to isolate the signal bandwidth, and may include realtime triggering to reduce the data volume by not recording signal-free data. The choice of whether or not to use triggering will be determined by the average fraction of time without detectable signals present in the data, the cost of data storage, and the difficulty of developing a reliable triggering algorithm in advance of recording the data. Detection systems typically fall into one of two cases: 1) there is a significant fraction of the data without detectable signals present, and data-storage costs are high enough that it's worthwhile to use a trigger to reduce the stored data volume, or 2) the fraction of data without detectable signals is small enough and/or the cost of storing the data is low enough that the effort to implement a trigger and the effect it has on the efficiency are not worthwhile.

5.4.3 Analysis of Alternatives

Triggering We have a choice for whether or not we use triggering for the LFA. In previous phases of Project 8, the higher cyclotron frequency of ≈ 26 GHz required the use of analog mixing to reduce the frequency band prior to digitization. The bandwidth of the frequency region of interest was ≈ 250 MHz. The data rate, for 1 B integers, recording IQ data, was $2 \times 1 \times 250 \times 10^6 = 500 \text{ MB s}^{-1}$. In Phase II we also had a relatively slow event rate, meaning that much of the data recorded in streaming mode would be pure noise. Therefore triggering was important.

For the LFA, with a higher data rate, there will be little noise-only time. The desired bandwidth is also much smaller, leading to a lower streaming data rate: $2 \times 1 \times 2 \times 10^6 = 4 \text{ MB s}^{-1}$.

ROACH2 from Phase II In Phase II of Project 8, we used a ROACH2 board [ref] as the main DAQ platform. While not directly associated with the development of the RFSoc, the ROACH2 had many similar capabilities as the modern AMD RFSoc platforms [ref]. The ROACH2, and its sibling platforms developed by the CASPER collaboration [ref], were developed prior to the existence of commercial RFSoc platforms, addressing the needs of astrophysics collaborations at the time. However, those needs can now be satisfied with RFSoc boards that are more capable than the older CASPER boards.

It would be possible to use the ROACH2 for the LFA. However, a newer RFSoc would be more reliable, and not require the use of older software. Therefore the RFSoc is definitely the preferred option.

5.5 CRES Calibrations

Primary Author: Stephen

In order to maximize the LFA's neutrino mass sensitivity, the statistical and systematic error terms described in equation 1 must be minimized. This requires a calibrations subsystem which can characterize the instrumental response to CRES electrons with high resolution (the calibration must be sensitive to small instrumental broadening terms σ_i) as well as high accuracy (the calibration must have small uncertainty terms $\delta\sigma_i$). The main

calibration methods will consist of direct CRES calibrations and indirect magnetic field calibrations. Additional diagnostic measurements (including microwave calibrations, tritium flux measurements, thermometry, and background magnetic field monitoring) allow for data quality cuts and quantitative modeling of the LFA's various subsystems.

5.5.1 Subsystem Objectives

Direct Calibration Sources In the direct CRES calibration measurements, mono-energetic electron sources are introduced in the LFA cavity, inducing CRES events with energies near the tritium endpoint. These calibration events closely match tritium events, providing realistic CRES signals and allowing for an end-to-end verification of the electron trapping efficiency, detection efficiency, and track reconstruction algorithms. In order to meet the LFA's goals, the linewidth of the electron sources must be narrower than $XXX\text{meV}$, and the absolute energies of the electrons must be known to $YYY\text{meV}$ or better. In order to capture calibration spectra without substantially reducing the live-time for science data, the electron sources must also generate CRES calibration events at an overall rate of several counts per second and be compatible with realistic LFA operating conditions (temperature, pressure).

Magnetic Trap Calibration The indirect magnetic field calibration maps the absolute magnitude of the magnetic trap. The known magnetic field strength is combined with an accurate measurement of the frequency of the CRES signal to reconstruct the electron energy with high precision and accuracy. In order to meet the LFA sensitivity goals, the magnetic field calibration must achieve a precision of $XXX\text{ppm}$ and an accuracy of $YYY\text{ppm}$. In order to minimize the live-time penalty, the magnetic calibration should also either be capable of achieving the required precision and switching back to a science data mode in a short time, or be capable of running simultaneously with tritium CRES events.

Diagnostic Measurements Additional diagnostic measurements are needed to thoroughly characterize the other LFA subsystems. In order to quantitatively model the full LFA apparatus and ensure reliable data quality cuts, the calibration subsystem will include microwave calibrations, tritium source calibrations, lithium beam calibrations, temperature measurements, and magnetic field monitoring. These diagnostic measurements must be sensitive enough to detect when the LFA is operating far from nominal conditions, allowing for data quality cuts. The diagnostic measurements should also allow for a quantitative agreement between modeled CRES events and real events, in order to provide a further cross-check on the main calibration systems described above.

5.5.2 Operational Principles

Direct Calibration Sources Introducing mono-energetic electron sources into the LFA is the most direct calibration method, as the electrons are trapped in the same apparatus and emit realistic CRES signals, which are handled by all LFA subsystems (cold electronics, warm electronics, triggering and analysis algorithms) in an identical manner to real tritium events. Several electron calibration sources are under consideration, including electrons generated by

radioactive decays as well as electron generated when a target gas is excited by an external source electron or X-ray source.

The gaseous electron sources will consist of metastable krypton-83 atoms filling the CRES cavity, which emit conversion electrons and gamma rays as they decay back to the ground state. The spectrum of conversion electrons is shown in figure 9, showing several narrow features with energies close to the tritium endpoint. A fraction of the emitted electrons have trappable pitch angles and generate detectable CRES events.

Figure 9: PLACEHOLDER: Spectrum of conversion electrons from metastable krypton-83, showing several available conversion electron lines with known energies and narrow linewidths.

Magnetic Trap Calibration

The magnetic field measurement is based on introducing particles with known gyromagnetic ratios (arising from nuclear or electronic spin), tipping them with RF pulses, and characterizing the frequency of the resulting Magnetic Resonance (MR) signal.

Diagnostic Measurements

5.5.3 Analysis of Alternatives

5.6 Magnetic Field - Integrated Design

Primary Author: Rene

The objective of an integrated magnetic field design is the combination and smooth working of all LFA magnetic field systems, including interaction and inter-dependencies between the different magnetic field systems. For CRES with an atom trap the main magnetic field contributions are:

- background magnetic field - sets the absolute cyclotron frequency
- CRES electron trap - to confine electrons within the detection region
- CRES atom trap - to avoid recombinations at the walls

In addition to these magnetic field several layers of correction systems are needed, e.g.

- magnetic shielding from the (lab) environment.
- a field correction system for spatial shimming and temporal stability.
- mapping device to map the in-situ magnetic field.

While we aim for a design as modular as possible and thus reduce risks and delays, many components impact others. The overall magnetic field design follows an onion like shell structure. The most outer layer is responsible to shield the experiment from (lab) environmental magnetic fields and changes, including the Earth magnetic field. The second layer provides the background field for CRES electrons. The electron CRES trap will be provided by the

same system as the background field. Further in a field monitoring system and shimming system will correct residual variations of the background and CRES trap magnetic field. Directly outside of the cavity the atom trap magnet is located. Only a small mapping device that can measure the magnetic field in-situ is placed in the active volume.

We discuss each subsystem in detail in [subsection 5.7-subsection 5.12](#). However, within an integrated magnetic field design we develop simulation tools to calculate the impact from one system on the other. This effects both mechanical stability and interacting forces by the magnetic field, as well as the effect on CRES physics parameters like efficiency.

The major interconnection and decision impacting all systems is the choice of the absolute background magnetic field which will be 560 MHz and the key aspects are discussed in [subsection 4.2](#).

5.7 Magnetic Field - Shielding from Environment

Primary Author: Rene

5.7.1 Subsystem Objectives

The objective of a magnetic field shielding layer from (lab)-environmental effects and changes is to compensate external magnetic fields and their variation to below $< \mu\text{T}$ and a frequency response of 1 Hz. Magnetic field changes at higher frequencies are effectively shielded by the skin depth effect in the aluminium of the cryo vessel.

External magnetic fields include the Earth magnetic field, since with 25–65 μT it is of similar scale as the background field strength of 21 μT . In addition magnet operations in neighboring laboratories, etc can cause transient magnetic field distorting the main background field.

Beside pure magnetic shielding, also an effective temperature stabilization and vibration mitigation is needed to keep running conditions inside the LFA stable. Temperature variation easily can lead to mechanical expansion on the level of $\sim 10^{-6}$ which in turn causes magnetic field variations on the ppm level.

Also vibrations of the mechanical setup displacing components by μm may cause magnetic field variations on the ppm level.

5.7.2 Operational Principles

The magnetic field correction and compensation system will consist of an active feedback system similar to n2EDM active magnetic shielded [8]. It consists of a set of 3-axis fluxgate magnetometers that monitor the magnetic field within the vicinity of the setup. An array of $\mathcal{O}(100)$ rectangular coils are covering the experimental hall walls that are used to compensate the external magnetic field. Shielding factors of ~ 15 can be reached for such a setup and frequencies < 0.1 Hz. The system allows for a high degree of flexibility. The system can be tailored to the experimental area local conditions. The control of $\mathcal{O}(100)$ separate coils for such applications has been demonstrated before.

For temperature stability a sufficient air conditioning system is required. In addition temperature isolation through either a wooden room or insulation blankets can help to keep temperatures stable. In addition also cryogenic gasses and liquids have to be temperature controlled.

5.7.3 Analysis of Alternatives

Several alternatives to the baseline design are considered. A cosine theta coil system that can correct the earth magnetic field similar to KATRINs Earth field compensation system. This system has much fewer coils and simpler geometry, however only can compensate the magnetic field in the z-direction, the direction of the background magnetic field. Since for CRES the magnitude of the magnetic field is relevant, variations in the transversal directions have only small impact. However since the Earth magnetic field is of similar strength as the background magnetic field the transversal magnetic field components significantly would impact the magnetic field magnitude.

A mu metal shield may effectively shield from the external magnetic fields, however the stray field of the background magnetic field may significantly saturate mu metal and thus making a mu metal shield non-functional.

A third option are super-conducting shields. However the presence of several other magnetic field components inside the shield would cause a storage of enormous amounts of energy in the coupled system and strong mechanical stress. In addition the freeze-in of a homogeneous magnetic field at may not be realizable.

5.8 Magnetic Field - Background Magnet

Primary Author: Hamish

5.8.1 Subsystem Objectives

The main magnetic field in which CRES spectroscopy is performed is a vertical homogeneous background field of 20 mT, corresponding to the chosen cavity frequency of 560 MHz. Additional magnetic fields provide electron and atomic trapping. The resolution requirements dictate that the azimuthal uniformity is $< 10^{-7}$ because azimuth is not a derivable parameter in TE011 cavity readout. A field standard deviation of 10^{-7} corresponds to an energy standard deviation of 53 meV. Temporal stability at this level is also required. The radial location of an electron can be determined at the 10% level or better from the coupled power, and the radial uniformity specification is therefore $< 10^{-6}$. Axial uniformity is less stringent still since the CRES configuration is in any case an axially non-uniform trap for electrons. Notwithstanding that, the uniformity in the orthogonal components determines also the axial uniformity, from Maxwell's equations. The addition of the electron trap introduces an inherent radial non-uniformity as discussed below. The mean CRES field for electrons undergoing axial oscillations in a trap can be precisely determined from the axial frequency as revealed in the sideband splitting. A prior calibration with the electron gun is required to relate the mean CRES field to the axial frequency.

These specifications are demanding, well beyond what is readily achievable, but we have reached a design combining mechanical tolerances with magnetic and thermal corrections that we consider can meet them.

5.8.2 Operational Principles

A short solenoid is not uniform but can be made so if the current density is allowed to vary along its length. This can be achieved in several ways; we elect to divide the solenoid into a stack of equally-spaced identical coils each carrying an appropriate, slightly different current to achieve uniformity. This choice has practical advantages in both manufacture and versatility of use.

If in a solenoid the magnetic field is perfectly uniform along the axis, then it is also perfectly uniform off the axis until one reaches current-carrying conductors. This follows from the divergenceless character of the magnetic field.

Consider a solenoid composed of n filamentary loops equally spaced by a distance a and all of the same radius R . Each loop i can be driven by an independent current U_i . The magnitude of the on-axis field of a single loop positioned at z_0 is:

$$B(z) = \frac{\mu_0 R^2 U}{2[(z - z_0)^2 + R^2]^{3/2}}. \quad (9)$$

For n such loops equally spaced by a , then, the field at a point z_i is

$$B_i = \frac{\mu_0 R^2}{2} \sum_{j=1,n} \frac{U_j}{[(aj - z_i)^2 + R^2]^{3/2}}. \quad (10)$$

The field in the useful region can be optimized by a least-squares method. This allows for some deviations within a desired range, and allows for optimization over an arbitrary number of test points along the axis. One may also weight the deviations for how important they are considered to be—for example, the central region may be more critical than near the ends. The equations contain the fit parameters (the currents U_j) and the desired data values (the constant field value $B_i = B$):

$$\begin{aligned} M_{11}U_1 + M_{12}U_2 \dots M_{1n}U_n &= B_i \\ M_{21}U_1 + M_{22}U_2 \dots M_{2n}U_n &= B_i \\ &\dots \\ M_{m1}U_1 + M_{m2}U_2 \dots M_{mn}U_n &= B_i \end{aligned} \quad (11)$$

where

$$M_{ij} = \frac{\mu_0 R^2}{2} [(aj - z_i)^2 + R^2]^{-3/2}. \quad (12)$$

There are m equations, one for each test point z_i , with $m \geq n$.

In [Figure 10](#) are shown the results from a fit for a magnet with 101 coils 1400 mm in diameter and equally spaced by 50 mm, a shorter solenoid than the 5.5-m one planned. Radial

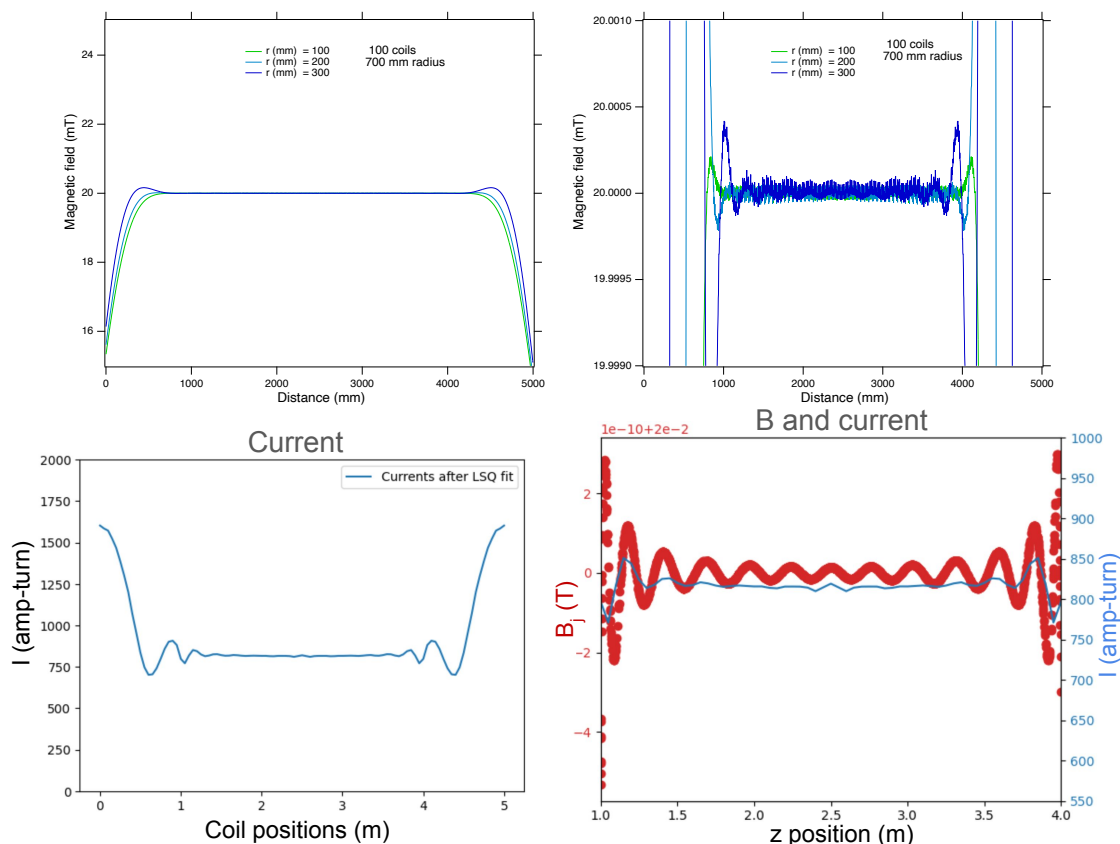


Figure 10: (Top left) Magnitude of the magnetic field for 3 radial displacements from the axis, 100, 200, and 300 mm. (Top right) Expanded view. (Bottom left) Coil ampere-turns. (Bottom right) Residuals from 20 mT on axis at 200 randomly distributed points, with the scale 2×10^{-10} T per division. The coil currents are superimposed (solid blue line), showing that the periodicities of the coil currents and residuals are not quite the same.

uniformity is confirmed at roughly the 1 ppm level, although the limits of numerical precision in this calculation are being approached. The structure of such a magnet is indicated in [Figure 11](#).

A magnetic ‘return’ is planned. It consists of large-diameter coils surrounding the main magnet and oppositely energized so as to confine the external field of the main magnet within the larger diameter. Magnetic returns reduce perturbations caused by the susceptibility of materials outside the working volume, and will also be needed in Phase IV of Project 8 where many cavities reside as neighbors. The return will modestly increase the required coil currents in the main magnet.

[Table 4](#) shows coil and total magnet parameters. The magnet coils are connected in series and energized by 112 independent power supplies. The primary current supply is to be stabilized by NMR or ESR free-induction decay probes positioned between the magnet and the cryostat to achieve the necessary 10^{-7} stability. Each individual coil is set to its exact current with a current-source supply. Powering the coils in this way is convenient

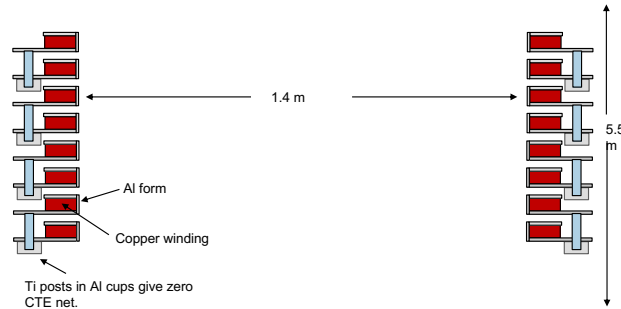


Figure 11: Part of a coil stack comprising the 20-mT background field solenoid (water cooling not shown).

and flexible, and leads to the simplest possible (and therefore lowest cost) coil. If the coils have a single winding, the number of turns determines the voltage for a specified number of ampere-turns, which cannot exceed the tolerance of the topmost supply for ground isolation. (An alternative is to include separate secondary windings on each coil to be powered close to ground potential.) The stability of the low-current trim supplies is less stringent, about the 10^{-5} level that is commercially available.

5.8.3 Imperfections

Thermal effects: A uniform-field solenoid is mostly immune to variations in the radius of coils, whether from mechanical variances or thermal expansion. However, the field is proportional to the linear current density, and therefore the overall length of the solenoid. The spacing of the coils will be made independent of temperature by use of compensated spacers consisting of a titanium rod (Grade 5) and an aluminum cup (T6061). Compensation at 25 C for a 50-mm spacing is achieved with a Ti length of 75.9 mm and an Al cup depth of 25.9 mm. The NMR/ESR stabilizers assure compensation at the next level.

Mechanical tolerances: The most serious imperfection is a curvature in the solenoid, a ‘banana’ distortion, because it introduces an azimuthal asymmetry. The solenoid becomes a section of a torus with the field larger on the small-radius side and conversely. To illustrate quantitatively, the overall solenoid length must be uniform azimuthally to within a wavelength of light. Such precision cannot be achieved by mechanical means, and correction coils consisting of vertical conductors on the exterior of the cryostat or the solenoid will be necessary. The conductors produce a non-uniform transverse field that adds in quadrature with the solenoid field to provide azimuthal correction of the field magnitude. A good mechanical baseline will be accomplished with precision-ground Ti-Al spacers individually measured by optical means and randomized or otherwise selected at installation, and the aggregate azimuthal variation is expected to be $< 10^{-5}$. Table 5 summarizes mechanical tolerances for a solenoid that meets specified field performance.

Ground motion: Microseismic, tidal, aeolian, and human-caused ground motion could excite swaying motion in a firmly anchored solenoid. It should therefore be mounted on a compliant suspension that provides isolation from external periodic forcing. Water-cooling

Parameter	Value (copper)	Aluminum	Unit
Field	20		mT
Height	5550		mm
Radius	700		mm
Coils	111		
Spacing	50		mm
Current (midpoint)	820		A-turns
Coil thickness	30		mm
Coil radial span	100		mm
Fill factor	0.6		
Effective cross section	1800		mm ²
Mean length	4712		mm/coil
Maximum total voltage	600		V
Resistivity	1.68×10^{-8}	2.65×10^{-8}	ohm-m
Resistance per coil	4.88×10^{-3}	7.70×10^{-3}	ohms
Power	3280	5200	W
Maximum turns per coil	149	95	
Density	8850	2700	kg m ⁻³
Mass (conductor only)	8333	2542	kg

Table 4: Power requirements for main magnet, default conductor Cu. (The power for individual coils near the ends is up to 4 times higher than the midpoint values in the table.)

turbulence and fluctuating supply pressure also must be avoided by using gravity feed from a reservoir or similar measures.

5.8.4 Analysis of Alternatives

Superconducting solenoids routinely produce magnetic fields uniform at the ppm level, particularly for MRI applications. With persistent switches, outstanding time stability can be relied on. Temperature stability is also not a concern. Nevertheless,

- A large superconducting magnet of this scale, even though the field is low, would be costly in comparison to the conventional magnet.

Parameter	Value	Tolerance	Unit
Solenoid length, azimuthal variation	5550	± 0.0005	mm
Flatness of individual coils	-	± 0.05	mm
Radius of individual coils	700	± 1	mm
Circularity of individual coils	-	± 1	mm
Thermal expansion of solenoid	-	± 1	ppm C ⁻¹

Table 5: Tolerances on dimensions if magnet uniformity is achieved by mechanical precision alone (provisional).

- It is technically difficult to bring out such a large number of leads from the cold bore.
- With many persistent switch settings, changes in the field become an elaborate procedure.
- Failure in any aspect of the magnet becomes a very serious matter to remedy.
- Procurement, manufacture, shipping, assembly and installation of a monolithic magnet of this size is a major undertaking.

Some consideration was given to adding a solenoidal winding to the Ioffe magnet, which is already at 4 K in a cryostat, but the above concerns remain. In addition, there are important advantages to being able to place an array of NMR/ESR monitors and other instrumentation inside the magnet but outside the cryostat. The periodic warmup for recycling tritium becomes traumatic for the magnetic system because the magnet must be shut down and re-energized each time. It is unlikely this could be accomplished without recalibration, significantly reducing live time.

5.9 Magnetic Field - Field Correction System

Primary Author: Rene

5.9.1 Subsystem Objectives

NMR probes in background magnet but outside ioffe magnet, in gap between two systems, monitor field , shim background field inhomogenities to X ppm

5.9.2 Operational Principles

individual coils that can be controlled to correct imperfections in background field, Ioffe field fast decaying thus no need for corrections, CRES trap relatively weak so no correction on level as background field Coils on zylinder surface, individually controlable, integrate with CRES coil possible

5.9.3 Analysis of Alternatives

5.10 Magnetic Field - Electron Trap for CRES

Primary Author: Rene

5.10.1 Subsystem Objectives

Trapping field to trap CRES electrons trappable pitch angle field shape axial frequency , cyclotron frequency shift double valued issues

5.10.2 Operational Principles

Cylinder with current loops, all current loops individually tunable, using Crawford method to define current per coil, cylinder symmetric, close to ioffe so no strong power loss

5.10.3 Analysis of Alternatives

5.11 Magnetic Field - Atom Trap

Primary Author: Hamish

Free atomic tritium is the source that Project 8 plans to use for measuring neutrino mass because it has a simple structure that offers an intrinsically narrow instrumental line width and freedom from theoretical uncertainties. Another advantage is that the electron scattering cross section per atom is about one-half that for the molecule. An atomic source then has about twice the activity of a molecular one for the same track length. These advantages are summarized in Table 6. The zero-point energy of the vibrational states of the tritium

Parameter	T ₂	T	Unit
FSD σ	433	0	meV
Doppler σ	162	1	meV
Cavity temperature	85	4	K
e ⁻ cross section per atom	1.8	0.9	10 ⁻²² m ²

Table 6: Comparison of some key parameters for atomic and molecular tritium. FSD is the Final-State Distribution of the molecule [9].

molecule sets a practical limit to neutrino mass measurements with the molecule. With Doppler broadening, the total width is about 462 meV rms [10]. If an atomic source can be made and operated at very low temperature, this spread can be essentially eliminated. The rms energy broadening is

$$\sigma(E)_{\text{Doppler}} = 2E_0 \sqrt{\frac{k_B T}{mc^2 \beta^2}}. \quad (13)$$

For molecular T₂ at 30K, it is 100 meV, and for atomic T at 1 mK it is 1 meV. Figure 12 compares the broadening contributed to the beta spectrum by the ground-state manifold for the molecular final states and by the atom.

Because atomic tritium (T) is reactive and tends to recombine with other tritium atoms, it must be magnetically confined and kept away from surfaces. Even superfluid He, which is usable as a container wall surface for atomic H, is not usable for T because the adsorption energy on He is too large [12]. In the gaseous state two-body recombination is not allowed and only three-body recombination can occur. Fortunately, this process is negligible at the densities of interest. The optimum density depends somewhat on experimental considerations, but is about $5 \times 10^{16} \text{ m}^{-3}$. A continuous supply of atoms is necessary to replenish the source against other losses.

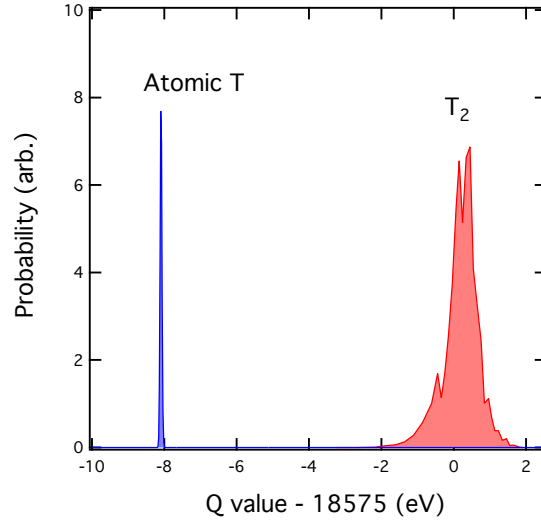


Figure 12: Comparison of the widths of the atomic and molecular spectra that broaden the beta spectrum, for the electronic ground states. Electronic excited states begin at 40 eV above the ground state for the atom and 20 eV for the molecule. Doppler broadening is shown for the atom at 1K to make it visible. The molecular spectrum is from Saenz *et al.* [11] (Doppler broadening for the molecule is omitted to illustrate the FSD).

Electrons are also magnetically trapped in a CRES experiment, but quantitatively the magnetic trap depth suitable for electrons is much smaller than that for atoms, unless impractically low atomic temperatures are reached. The two constraints can be decoupled by using a magnetogravitational trap in which the magnetic boundary at the top is the right strength for electron trapping while the trap’s height is sufficient to prevent atoms from escaping. Pioneering work on magnetogravitational traps has been carried out for measuring the lifetime of the neutron [13,14]. The temperature scales for gravitational and magnetic trapping are, respectively,

$$k_B T = mgh \quad (14)$$

$$k_B T = \mu \cdot \mathbf{B}. \quad (15)$$

Numerically (for T), these correspond to 3.54 mK/m and 671 mK/T. The magnetic field equivalent to 1 m height is 5.27 mT.

Two technologies are known for pure static magnetic trapping, the permanent-magnet Halbach array and the Ioffe-Pritchard trap, an array of conductors carrying currents in alternating directions so as to create a strong field near the array. Both can produce a field that falls off steeply toward the interior where the CRES spectroscopy takes place. Atoms are adiabatically reflected at the magnetic ‘wall’ while the uniform CRES field ideally remains undisturbed. The Ioffe method is selected for trapping T in the LFA.

A low physical temperature of surfaces is required for other reasons. As may be seen in Fig. 12, the endpoint energy of the T spectrum is about 10 eV less than that of the T₂ spectrum [10]. Any molecules remaining in the trap represent a background at the T

endpoint, and must be kept to less than $\sim 10^{-4}$ abundance. The saturated vapor pressure of T₂ and HT over the solid is given by Souers *et al.* [15] as an empirical equation,

$$\ln Q = A + \frac{B}{T} + B' \ln T \quad (16)$$

and is shown in Fig. 13, converted to mbar. The constants are given in Table 7. This calls

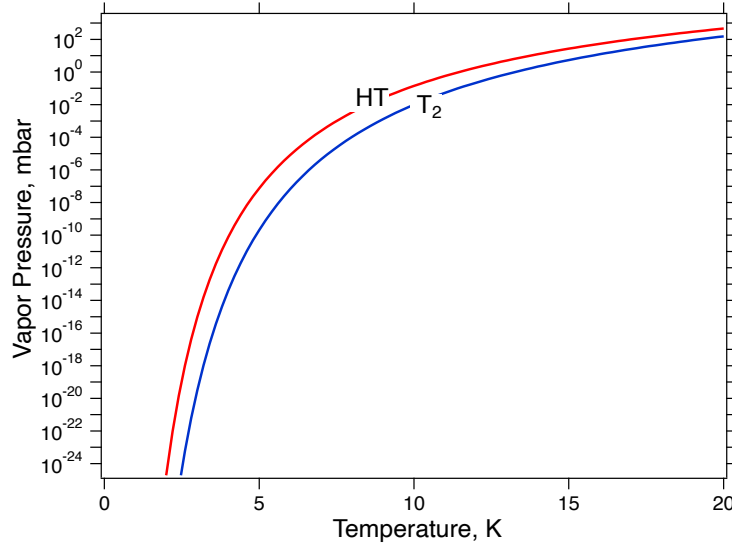


Figure 13: Saturated vapor pressure of T₂ and HT over the solids [15].

Constant	HT	T ₂	Unit
<i>A</i>	4.559165	5.846049	Torr
<i>B</i>	-126.6	-160.7	Torr-K
<i>B'</i>	2.552	2.3235	Torr

Table 7: Vapor pressure constants for tritium isotopologues [15].

for a cavity with physical walls at a temperature $\lesssim 3.5$ K to assure negligible molecular contamination. The molecule, having no appreciable magnetic moment, is not trapped, strikes a wall, and is frozen out there with negligible vapor pressure. Desorption of tritium induced by beta decay in the deposited layer [16] is a separate source of contamination that limits the amount deposited on the wall before the wall must be warmed to pump away and recycle the tritium accumulated there. The low cavity temperature also is valuable in reducing noise and improving the SNR.

Atomic tritium is a simple quantum system in which the total angular momentum is formed from the nuclear and atomic spins, each $1/2$. The Breit-Rabi diagram is shown in Fig. 14. The 4 hyperfine states are conventionally named *a*, *b*, *c*, *d* as shown: the low-field seekers are *c* and *d*, collectively T_↑, whereas the high-field seekers are *a* and *b*, collectively T_↓. At fields above about 0.1 T, the spins orient more or less independently, defining a

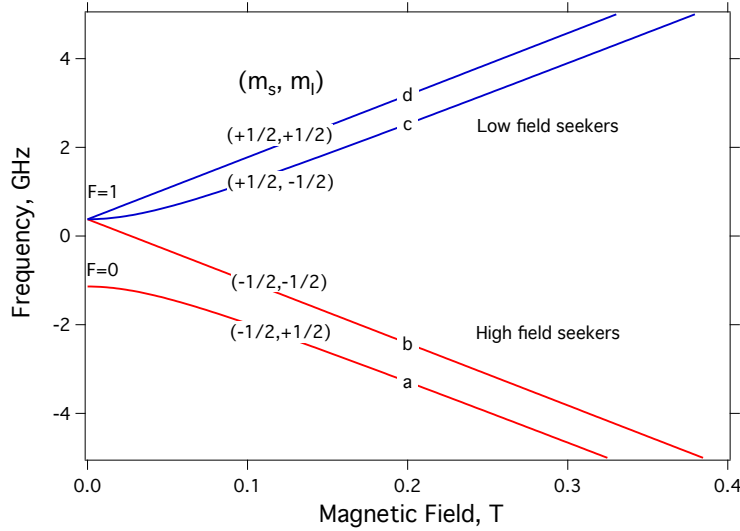


Figure 14: Breit-Rabi level diagram for atomic tritium. The level energies are from Silvera and Walraven [12].

‘high-field’ region, but states b and d are pure spin states with the same magnetic moment, approximately 1 Bohr magneton, at all fields.

5.11.1 Subsystem Objectives

The LFA is the first large-scale tritium experiment to be usable with atomic tritium. It is large enough to test all the principles of the final Phase IV experiment and to be capable of delivering a measurement of neutrino mass at the frontier established by KATRIN.

The magnetic design of the atomic trap must have a magnetic ‘wall’ height at the physical wall of the cavity that is large enough to reflect low-field-seeking atoms with high efficiency back into the central volume of the cavity without allowing them to reach the cavity wall. The magnetic wall must nevertheless have negligible effect on the uniform central field where CRES spectroscopy takes place. A steep spatial decline from the wall is therefore required. The magnetic wall must be free of low spots where atoms could escape and it must not create magnetic zeroes where Majorana spin-flips could occur, requirements that are similar but not the same. The height of the magnetic wall is tied to the overall physical height of the cavity – losses of hotter atoms at the top sets a temperature limit for the trapped atoms, and the magnetic wall should contain those same atoms without contributing significant additional losses.

The design must meet external constraints including injection of the continuous supply of cold atoms, compatibility with operation below 4 K and the periodic recycling of cryosorbed T_2 , and long-term stability and reproducibility.

5.11.2 Design

A superconducting (SC) Ioffe trap has been designed in collaboration with Vikas Teotia of the Magnet Division of Brookhaven National Lab. The dimensions needed by Project 8 for the LFA are a match to a direct-winding machine, the largest in the world, built at BNL for EIC magnets. The Ioffe conductor locus calculated by Project 8 is shown in [Figure 15](#). It is a

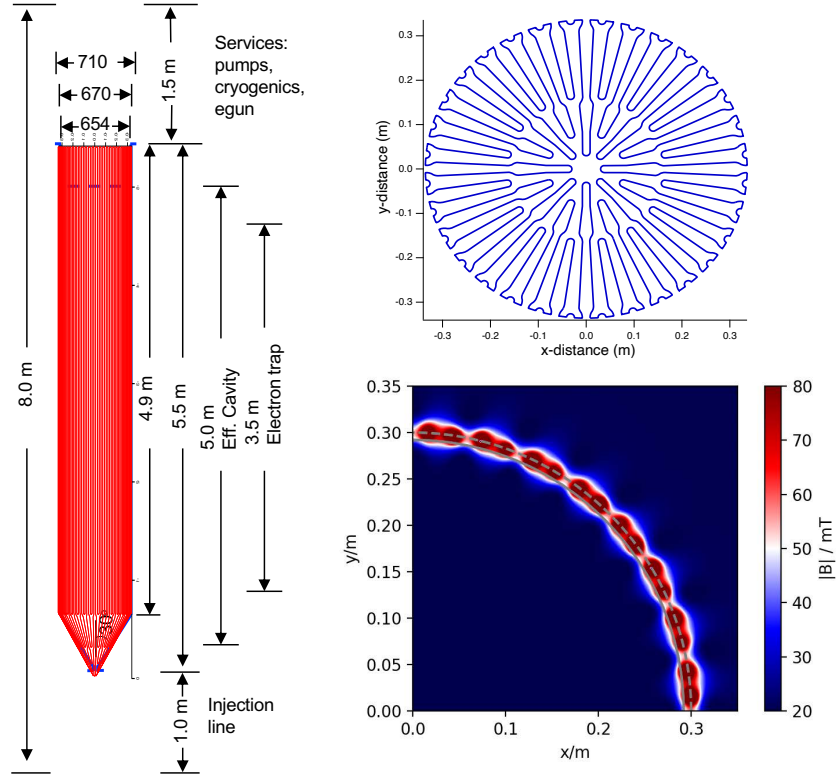


Figure 15: Design for superconducting Ioffe trap for the LFA. The left panel shows the 64-pole magnet wound on the exterior surface of the cavity. The magnetic closure at the bottom is conical to avoid field zeroes. Dimensions are only indicative. The top right panel is an end-on view of the locus on the cone. The lower right panel shows the calculated field magnitude part way down the cone. The dotted line is the Ioffe conductor position and the solid line is the cavity wall.

64-pole multipole (128 parallel conductors on the surface of the cylinder). Multipolarities m that are powers of 2 have no residual transverse field components. The field declines radially inward as $(r/r_{\text{cavity}})^{m/2-1}$. A 32-pole magnet produces a field that falls off too slowly, limiting the usable volume of the cavity. With a 128-pole magnet the turning radii at the ends of the loops on a 560-MHz cavity are too tight for standard SC cable and the field declines so quickly it becomes too small for practical cavity wall thicknesses.

The cavity is aluminum. The default superconductor choice is NbTi strands 40% by volume in a 1.45-mm diameter copper matrix epoxied into precision-milled slots. The number

of conductors required to reach the desired 50-mT field is not yet determined. To withstand collapse against atmospheric pressure (not a normal condition in operation) the wall thickness must be at least 6.24 mm from cavity wall to the slot nadir.

5.11.3 Imperfections

As with the main magnet, the most serious imperfection is azimuthal errors. Azimuthal displacement of a conductor creates a non-uniform transverse field component. The slots for conductors on the cylinder must be azimuthally located to within $\pm 100 \mu\text{rad}$. Errors in the radial coordinates of conductors also produce azimuthal fields. **The tolerance is being evaluated.**

5.11.4 Analysis of Alternatives

The alternative approach to this design is a Halbach array of permanent magnets. The reasons for choosing the superconducting Ioffe coil are,

- Greater precision in cancellation of the interior field is probable with the Ioffe. The magnetic material, a sintered powder, exhibits small variations in composition, density, dimensions, and magnetization. At the level of precision needed, there is a lack of detailed analysis methods for a realistic Halbach array (tolerances, variability of magnets).
- The Ioffe trap strength can be adjusted or set to 0.
- The cavity operates at $\sim 4 \text{ K}$ in any case, lowering incremental cost for SC technology.
- There is less mass to warm up to 20 K for periodic T_2 recovery.
- There is less mass to suspend and cool in the 4 K cryostat.
- Permanent magnets have a non-zero residual permeability (1.05 for NdFeB), disturbing axial field uniformity in the interior and introducing possible hysteresis.
- Less dependence on a foreign supply chain is anticipated.

We conclude that a practical and cost-effective design for a large Ioffe trap has been identified, one that meets the needs of Project 8 in conducting a future atomic tritium experiment with the LFA. The trap in the LFA will first be used with Li atoms for diagnostic evaluation.

5.12 Magnetic Field - Mapping

Primary Author: Rene, Elise

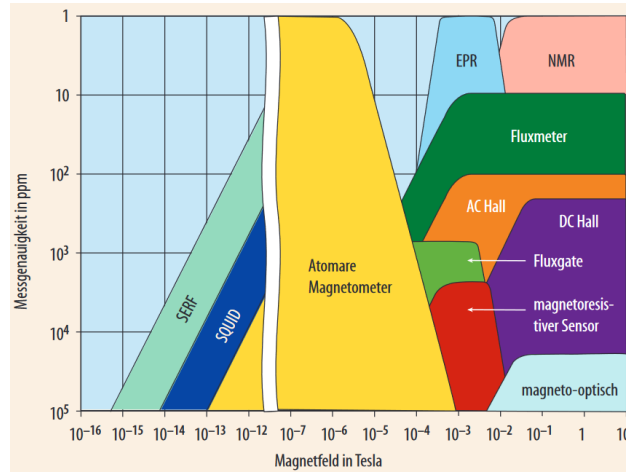


Figure 16: Overview of typical accuracy attainable with different magnetic measurement methods depending on the measured field. Note the jump by 5 orders of magnitude at the lower end of the magnetic field scale.

5.12.1 Subsystem Objectives

While a careful design of magnetic field systems and its simulation are essential to produce a homogeneous magnetic field for CRES, the systematic uncertainty depends on the accuracy and precision with which the magnetic field is known and thus can be measured. Residual imperfections have to be mapped in-situ since any material adds magnetic field perturbations. Therefore, a mapping system of the magnetic field inside the cavity volume is necessary. The system should be operated at cryogenic temperatures, while no gas is filled in the cavity. The low temperature, very limited excess and low magnetic field as well as the high-precision and accuracy requirement make this task particularly difficult. The mapper consists of one magnetometer that is able to measure magnetic fields at $< 10^{-7}$ level at 20 mT fields and is cryo compatible. In addition a positioning system of this magnetometer with spatial precision of $< \text{mm}$ is needed which can measure the volume at X positions within X time.

5.12.2 Operational Principles

Classical magnetometers that reach sub-ppm precision are ESR/EPR and NMR magnetometers. The magnetic field of 20 mT lays exactly in the transition region between these two systems. Table 8 and Table 9 give the gyromagnetic ratio γ for NMR nuclei and the g -factor for some ESR molecules, respectively. For an absolute magnetic field measurement which is accurate, the gyromagnetic ratio or the g -factor must be known to better than 100 ppb. None of the ESR molecules reaches this accuracy and only ^1H and ^3He gyromagnetic ratios are known to this accuracy.

Both ESR and NMR magnetometer measure the Larmor frequency of the electron and nuclear spin precession signal, respectively. The magnetic field is determined from the Larmor frequency and its resolution depends on the frequency resolution. For gaussian distributed

Nucleus	$\gamma_n/(2\pi)$ [MHz T ⁻¹]	rel. accuracy
¹ H (free)	42.577 478 461(18)	7.0 ppb
¹ H (in H ₂ O)	42.576 385 43(17)	4.0 ppb
² H	6.536	
³ H	45.415	
³ He	−32.434 100 033(28)	0.86 ppb
⁷ Li	16.546	
¹⁹ F	40.078	
<i>e</i> (free)	28 024.951 386 1	

Table 8: Gyro-magnetic ratio for several nuclei and a free electron.

noise a lower limit on the magnetic field resolution can be calculated using the Cramer-Rao lower bound [17]

$$\frac{\Delta B}{B} = \frac{\sqrt{\sigma_{f_L}^2}}{f_L} \geq \frac{\sqrt{12}}{2\pi \cdot SNR \cdot \sqrt{f_{BW}} \cdot t_{\text{obs}}^{3/2}} \sqrt{C(t_{\text{obs}}, T_2^*)} \frac{1}{f_L}, \quad (17)$$

where $f_L = \frac{\omega_L}{2\pi} = \gamma B$ is the precession frequency, t_{obs} the length of the measurement time, $f_{BW} = sr/2$ is the sampling rate limited bandwidth, SNR is the signal-to-noise ratio and C is a numerical constant of order $\mathcal{O}(1)$ that depends on the decoherence time T_2^* .

Gaseous magnetometers profit from motional-averaging over the sample volume and are less effected by magnetic field gradients. This increases the coherence time T_2^* allowing for longer observations. Gaseous magnetometers usually have smaller number densities compared to solid-state magnetometers, however high-pressure cells can partly compensate this drawback. At to high pressures however the advantage of motional-averaging can be lost again. For gaseous magnetometers in a motional-averaging regime, the signal is a exponential damped sine signal.

The difference between ESR and NMR is the gyromagnetic ratio γ and different sensitivity to shielding effects which depend on the environment of the atoms or the nucleus.

The SNR depends on the type of magnetometer and the temperature of the first amplifier. For hyperpolarized gases where a polarization close to 100% is reached, the SNR depends linear on the magnetic field. For thermal polarized gases the degree of polarization scales like B/T and thus the SNR depends on B^2/T . An additional $1/\sqrt{T}$ factor may be due to the amplifiers temperature.

$$\omega_{\text{Lamor}} = \gamma B \quad (18)$$

$$= g \frac{\mu_B}{\hbar} B \quad (19)$$

$$g = g_e \cdot (1 - \sigma) \quad (20)$$

³He hyperpolarized sample on 3D movable arm

Substance	g-factor	rel. accuracy	Reference
free electron (e)	2.002 319 304 361 18(26)	0.13 ppt	Phys. Rev. Lett. 130 (2013) 157501
charred dextrose (CD)	2.002 592 2(25)	1.2 ppm	J. Magn. Reson. 6 (1973) 100
2,2-diphenyl-1-picrylhydrazyl (DPPH)	2.0036(2)	100 ppm	Phys. Rev. 77 (1950) 1405
1,3-Bisdiphenylene-2-Phenyl Allyl (BDPA)	2.0026(2)	100 ppm	J. Chem. Phys. 39 (1963) 1555
nitride-oxide (NO)			
LiF:Li	2.002 293(2)	1.0 ppm	Phys. Lett. A. (1989) 131
(H@iBuT ₈)	2.002 94(3)	15 ppm	J Magn Reson. 2010 Aug 1

Table 9: ESR/EPR g -factor measurements of several molecules.

5.12.3 Analysis of Alternatives

Electron-Spin-Resonance / Electron-Paramagnetic Resonance Phased ESR

Rydberg atoms

other NMR

5.13 Lithium Beamline

Primary Author: Ben Jones

5.13.1 Subsystem Objectives

The lithium beamline delivers lithium atoms into the atomic trap and is used to characterize the trap lifetime and prove that performance of the CRES system is maintained in the presence of the magnetically trapped atomic cloud. Electrons will be introduced into the CRES cavity by scattering off stored, spin-polarized lithium leading to a CRES source spanning full relevant phase space of resolvable electron trajectories within the active volume. The lithium will be monitored by laser diagnostics at 670 nm both along the beamline and inside the trap to characterize performance. Spatially resolved maps of temperature, atomic density and velocity will be made to study the dynamics of the trap loading and thermal equilibration, important performance parameters for the Project 8 atomic concept.

5.13.2 Operational Principles

A lithium source is currently being developed by Project 8 as a proof of principle for Magnetic Evaporative Cooling Beamline (MECB) scheme that will be implemented for tritium delivery in Phase III. Tritium cooling must proceed purely magnetically without magneto-optical elements due to the inaccessibility of the Lyman α line with sufficiently powerful lasers. For lithium however, magneto-optical methods are available, tuned on the D lines at 670 nm. To emulate the flux emerging from the tritium accomodator, the lithium R&D system uses a Zeeman slower with a detuned laser aimed up-beam of an atomic flux produced in a hot oven to pre-slow it to around 10 K. This technique can be used to slow lithium still further, to the point that the trap can be loaded without requiring MECB elements. Loading of atom traps directly from lithium Zeeman slowers has been demonstrated in the literature [citation(s)]

and we thus characterize this approach to trap loading as carrying very low risk. As MECB elements become available following our atomic cooling R&D they can be added into the beamline for the LFA in later runs, broadening the scope of the demonstration of the atomic beamline system and further increasing the impact of the LFA to retire risks associated with the tritium phase.

The atomic density needed for successful use of the electron gun to produce calibration electrons on ${}^6\text{Li}$ is [Need this number]. To produce the initial flux, isotopically enriched ${}^6\text{Li}$ will be heated to approximately 530 C to produce Li vapor. The isotope ${}^6\text{Li}$ is favored over ${}^7\text{Li}$ due to its larger triplet scattering length. Disappearance of the cross section for identical fermionic atoms at zero momentum is expected, but its effects are circumvented in our scheme by cooling a mixture of the low-field-seeking hyperfine states. The transport time in the beamline is insufficient for major losses to occur through relaxation, though it is anticipated that only the maximally stretched states will ultimately have a long trap lifetime.

From 530 C at the lithium oven, Zeeman slowing over a 2 m long beam-line with a counter-propagating, slightly de-tuned 670.992 nm laser and a slowly varying solenoidal field plus repumping at the 670.977 nm transition suffices to provide a cooled and almost stationary beam at the trap. The laser will be frequency locked on the partially cooled beam using a Doppler-free configuration following standard methods, and detuned via acousto-optic modulators (AOMs) to the relevant slowing and repumping frequencies. Thermometry will be performed through absorption measurements on the same 2S-2P transition by scanning over frequencies with an independent ECDL after the initial slowing stage, and spatial density mapping via fluorescence imaging using a scientific CCD camera viewing the trap volume. A permanent magnetic quadrupole channel in a bending configuration at the end of the beamline will divert the trapped atoms from the slowing portion into the trap volume vertically.

5.13.3 Analysis of Alternatives

An initial design concept considered the use of atomic hydrogen to load the trap for the first LFA phase. However, recent work on atomic cross sections has illuminated that elastic triplet scattering of H at low temperature has a cross section 10^5 times lower than T, suggesting that cooling and trapping of large H fluxes will be significantly more difficult. Direct demonstration with T, on the other hand, requires a large tritium throughput for successful thermalization that underlies the MECB dynamics, and it is not yet clear whether this can be demonstrated with the expected T inventory outside of a major tritium handling facility. The R&D work on H or T cooling systems in Project 8 is advancing rapidly, however, and we note the option for either an alternative or for an upgrade path for trapping trap H or T may provide an even more direct demonstration of coincident trapping and CRES measurements relevant to Project 8.

The Li delivery concept is based on a very low risk baseline, which is direct trap loading from the aperture of a Zeeman slower; as well as a higher risk stretch plan with more compelling physics deliverables, which involves installing MECB elements in the beamline and prove the ultimate Project 8 trap loading concept. The lithium beamline is a standalone component that can be decoupled and re-coupled to the trap at any time, and so the decision

about whether to augment the baseline plan can be made later, when further R&D results are in hand. Magnetic elements of the MECB are not expected to be cost drivers, and as such we do not expect this technical decision to substantially alter the lithium beam budget. We do intend to plan for acquiring enough vacuum pumps to evacuate the entire line within the baseline budget, however, to allow for this upgrade. This implies that the pure Zeeman system will be somewhat over-outfitted with vacuum pumping capacity.

Alternative alkali atomic sources are also possible [**Do we want to talk about Ben M's ideas for other atoms here?**] and can be delivered using the same beamline augmented with a different laser wavelength. These would be simple upgrades that can extend the science reach of the LFA for future runs. In-situ atomic magnetometry using trapped lithium or other alkali atoms is also a compelling R&D line that is under exploration, enabling LFA to serve as an R&D platform for the broader Project 8 atomic program.

5.14 Gas Handling

Primary Author: TBD

5.14.1 Subsystem Objectives

5.14.2 Operational Principles

5.14.3 Analysis of Alternatives

5.15 Cryogenics and Infrastructure

Primary Author: ?

5.15.1 Subsystem Objectives

5.15.2 Operational Principles

5.15.3 Analysis of Alternatives

5.16 Slow Controls

Primary Author: Walter Pettus

5.16.1 Subsystem Objectives

5.16.2 Operational Principles

5.16.3 Analysis of Alternatives

6 LFA Operations

Primary Authors: Elise, Noah, Brent

Describe the Operating cycle of the LFA. Major points to cover are:

- Description of scientific run modes (including calibrations)
- Description of maintenance modes
- Estimates of the operations cycle (how often do we do each run mode, and for how long).
- Description of data rates and volumes for each mode, and the management. What's the shipping and storage scheme, and what processing will we do to it.
- Operations Roles: shift operators, on-call experts, run scripting, data quality monitoring

LFA Run Modes: Describe the run/operation modes. We distinguish M modes of data taking and N maintenance modes. Details of how these modes work and what they accomplish should be already covered elsewhere. Focus here

Electron Gun Operation: Operations w/ e-gun on gases other than lithium.

Krypton Operations: Operations with krypton

Other calibrations: This is possibly multiple entries, depending on how section 5.5 develops.

Lithium Operations: The primary scientific operating mode of the LFA. Focus on lithium, but be aware that this eventually upgrades to tritium.

LFA Maintenance Modes:

Lithium/Tritium Recovery The time cycle for recovering lithium, and eventually tritium that has been lost throughout the system. The procedure is probably covered elsewhere.

Others?: possibly multiple entries

Computing and Data Management:. Feedback cycle of operations (data), analysis, and simulations.

Data Rates/Volumes: Cover each of the operation modes above

Simulations Data: A major body of data not covered by one of the operating modes

Collaboration Access: How/where will the data be stored and what's the model for sharing with all collaborators?]

Include tables and/or plots of data rates/volumes and the duty cycle of the run modes above.

7 Simulations

7.1 Existing work

The current simulation framework in Project 8 consists of two main software packages in addition to a suite of commercial and custom tools for targeted studies. Within the framework, particular emphasis is placed on accurately modeling the observable physics in CRES, while simultaneously maintaining compatibility with existing calculations. The Kassiopeia [18] package, developed for the KATRIN experiment, is a toolkit for particle tracking simulations in experiments containing complex geometries and electromagnetic fields. Within Project 8, electron trajectories are calculated for a given detector geometry and magnetic trap. Electrons with a range of initial energies, positions, and momenta are simulated. A second package developed by the Project 8 collaboration, Locust [19], is used to simulate the measurement of time-varying electromagnetic fields radiated by the Kassiopeia electrons. The software classes in Locust and Kassiopeia are tightly integrated, supporting bidirectional technical communication between the packages while the calculations are running. The resulting simulation framework has provided concreteness to theoretical models and has allowed for flexible integration of physical constraints that are specified with external or independent calculations. An example of the output of a CRES simulation is shown in Figure 17.

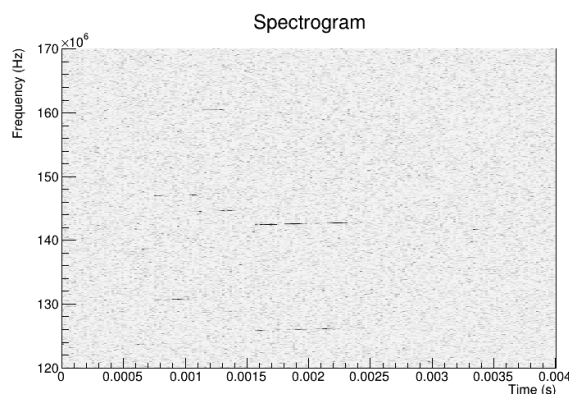


Figure 17: Simulated signals from three magnetically trapped 18.6 keV electrons with a background field of 0.95 T. Temporal discontinuities in the tracks are due to simulated interactions with hydrogen gas molecules, calculated with the Kassiopeia package.

Within the above configuration, Locust models the phase-sensitive RF response of the Project 8 experiment. It can include single or multiple resonant cavity modes with individually configured quality factors and resonant frequencies. The simulated electron's location and momentum relative to the mode fields determine its coupling to the mode. The resonant enhancement of each mode is calculated by convolving a time history of the electron's cyclotron frequency with a configurable finite impulse response filter. Locust can accommodate either analytic mode maps or empirical mode maps imported from a commercial package such as HFSS [20]. One or more configurable readout probes couple to the excited mode

based on its location. Finally, the simulated readouts from the probes incorporate effects from real experimental setups into the recorded time series, including down mixing with a local oscillator or a specified dynamic range for a data acquisition instrument.

7.2 AoA

While many commercial packages exist either to simulate time-domain RF circuits or solve EM fields, co-simulation packages to integrate both tasks have not been as widely developed. The main benefit of utilizing flexible and open source software packages like Kassiopeia and Locust is the ability to customize and integrate models of both problems simultaneously. But while Kassiopeia and Locust are the default simulation tools, other commercial products may still be viable as cross checks or as alternatives. Packages like CST [21] provide a potential option for a sophisticated cross check of electron trajectory calculations, though the cost for the software license as well as its computational speed are hurdles towards making this the main tool for Project 8 development. In addition, CST is closed-source software, which limits its compatibility with other custom libraries. Similarly other tools [22–24] can address the RF portion of the simulations, but like CST they are at least partially closed-source and integrating them with the EM field solvers would be non-trivial.

7.3 Goals and applications

With the simulation framework, quantitative studies can be performed on the simulated CRES data. Some examples of targeted studies are

- Reconstruction and trigger efficiency studies.
- Evaluation of design performance and tolerances.
- Evaluation of alternative experiment configurations.

Additional studies with broader focus could include

- Interpretation and analysis of laboratory data.
- Ongoing refinements to existing simulations as required by analysis goals.

7.4 Next steps

7.4.1 RF signal simulation

The Locust software package models the response of a detector readout to the time-varying RF field emitted by electrons modeled either analytically or in Kassiopeia. Previous phases of Project 8 utilized this framework to model RF responses to electrons in both free space and waveguides. For the current and future phases of Project 8, resonant cavities are the main technique being investigated. This motivates several ongoing goals for the Locust framework. First, the absolute normalization of the signal power in the cavity, including the quality factor and Purcell effect, requires additional development. The absolute signal power

with multiple mode excitations is also of interest. Developing these capabilities, as well as the ability to import multiple numerically calculated mode maps, will aid in the investigation of multimodal approaches to event reconstruction.

7.4.2 FFTM upgrade

The Kassiopeia/KEMField package is a toolkit that allows users to solve electrostatic and magnetostatic fields from user-defined inputs. Its primary application has been the KATRIN experiment, and as such the electric field calculations have been of critical interest. In Project 8, where magnetic trapping is prevalent, additional development on KEMField magnetic field calculations would likely be beneficial. For example, the present KEMField magnetic field calculation in a rotationally asymmetric geometry relies on a direct Biot-Savart calculation. While the latter approach is accurate and is useful in compact geometries, its nominal computation time scales upward with the number of current elements. As a result, for e.g. the rotationally asymmetric atomic trapping configurations in Section 5.11, exact calculations of CRES electron trajectories become expensive. A similar limitation was noted previously in the KATRIN experiment for electric field calculations. This challenge motivated a novel and efficient implementation of the Fast Fourier Transform on Multipoles (FFTM) method [25] for electric fields in KEMField.

In the FFTM implementation, detailed in [25], the electric potential $\phi(\mathbf{r})$ at location $\mathbf{r}$ is expressed in terms of derived multipole coefficients weighted with surface charges $\sigma(\mathbf{r}')$ on an electrode at $\mathbf{r}'$ with normal $\mathbf{n}$. In translating the calculation to magnetic fields, magnetic pseudo-charges $\sigma_m(\mathbf{r}') \equiv \mathbf{n} \cdot \mathbf{B}$ can in principle be applied in place of the electrical surface charges, assuming a magnetic scalar potential Φ_M such that $\mathbf{H} = -\nabla\Phi_M$ [26]. The outcome of the FFTM implementation for magnetic fields is expected to be a substantial increase in computational efficiency, allowing for detailed studies of electron motion in rotationally asymmetric magnetic traps.

7.4.3 Magnetic field map imports

As an important complement to the FFTM upgrade described above, imported magnetic field maps can provide flexible and relatively efficient calculations for geometries with arbitrarily configured current elements. Computations with rotationally asymmetric geometries, as are of continuing interest in Project 8, would naturally be well suited to this approach. In addition, empirically measured field maps resulting from the work detailed in Section 5.12 must also be imported into the simulation for quantitative interpretation. The relevant field maps can be measured or calculated over a grid of points using one of the discretized or integrating approaches available in Kassiopeia/KEMField, or can in principle be extracted from external packages such as Magpylib [27]. Following the import to Kassiopeia/KEMField, the magnetic field values can be calculated between the grid points by interpolation as required for the trajectory calculations. Where computational approaches differ in the field map derivations, such as with analytic calculations in Magpylib and discretized numerical algorithms in Kassiopeia/KEMField, valuable cross-checks between the packages can be applied in post-simulation analyses.

7.5 Commercial simulation packages

Within Project 8, focused technical questions often arise that are well-suited to highly developed commercial simulation packages. High Frequency Structure Simulation (HFSS) [20] is a sophisticated industry-standard tool offering a comprehensive set of solvers to address diverse electromagnetic (EM) problems. Figure 18 shows an example of an EM solution in Project 8 calculated with HFSS. The similarly advanced Computer Simulation Technology (CST) package provides an additional computational resource. Depending on the problem at hand, a user might prefer the I/O interface of one package, or for example might find the time-dependent electromagnetic solutions to be more readily accessible in an alternative solution. The Comsol Multiphysics package provides an additional platform to simulate coupled geometries, material properties, and EM physics solutions, and is accessible through a well-developed library of deployable examples. Finally, in work toward an atomic design concept for Project 8 described in Section 5.13, the Sparta [28] package has been a useful tool for performing direct Monte Carlo simulations of low-density gases in 2D or 3D. It should be noted that while each of the packages mentioned above can generate useful and independent technical information, the central Project 8 simulation framework has purposely been configured to incorporate the quantitative results of the commercial packages as stringently as possible.

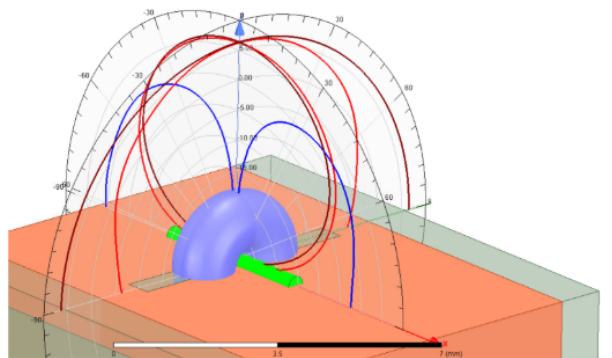


Figure 18: HFSS simulation of the field magnitude induced in a slotted rectangular waveguide driven with a sinusoidal voltage. Credit: T. Wendler

7.6 Outlook

Additional future developments within the Project 8 computational framework will be needed to study and quantify systematic effects that are relevant to the neutrino mass. As in previous work, new computational development must draw from or agree with existing calculations and measurements. Similarly, for meaningful interpretation, new calculations should in principle be compatible with existing Project 8 software libraries. Examples of future simulation capabilities should include plasma effects such as those in [29], in which numerical methods such as “particle-in-cell” are applied to nonthermal particle distributions. A careful approach

to a quantitative interface between the plasma physics and the existing Kassiopeia simulation will be needed. Additionally, updates to available particle interaction cross sections will likely be required, both for electron-ion collisions and neutral particle interactions.

Within the existing simulation framework, new applications of interest continue to be identified as the Project 8 experiment plan advances. For example a resonant cavity simulation in Kassiopeia/KEMField, where an electron in Kassiopeia excites cavity mode fields defined in KEMField, would be both interesting and useful for comparison with other analogous calculations. The currently available infrastructure could be readily utilized for this work. Other relevant configurations can in principle be built and used to test new components in the experiment. Such candidates could include an electron in a Penning Trap combined with synchrotron excitation techniques, conceptualized as an electron source in [30] and based on work in the field of relativistic electron trapping by G. Gabrielse and others [31]. A working simulation of the latter could define a path forward or deepen the understanding of its realization. Other candidate ideas will likely emerge as the Project 8 experiment plan unfolds, and as the data and computing infrastructure described in Section 6 is further developed.

8 Offline Data Reduction and Analysis

PTS: replaced with "Offline Data Reduction and Analysis"

Primary Author: Pranava and Noah

Track and event reconstruction, and spectrum analysis. Sensitivity will have been covered earlier. DAQ and warm electronics covers getting data onto disk; begin here with reading from disk.

Broadly speaking the offline processing and analysis includes (ordered in time):

1. Triggering: Binary selection of the physics data
2. Event detection: Within the triggered data, identification of the events; in the P-II lingo this includes everything from RF BG shape removal to the track segment clustering (in principle it also includes the event power cuts)
3. Event-by-Event reconstruction: Given an event, extraction of the individual event parameters including start frequency (and energy), axial frequency/extent, pitch angle, radial position, etc. There can be and (is for P-II) an overlap between detection and reconstruction
4. Spectrum analysis: Ensemble-level analyses; includes lineshape analysis (for calibration mainly) + continuum (for neutrino mass extraction)

At each step above, the performance of the algorithms is assessed. Further the detector instrumental response model is also developed by taking inputs from all the steps.

We discuss the above into three steps: Data processing, reconstruction, and spectrum analysis.

8.1 Data Processing

8.1.1 Triggering

While triggering played an important role in P-II and will play an important role in CCA, we estimate that the data rate is low enough for detectors operating at sub-GHz frequencies ala LFA, that we can stream data fully.

One paragraph on the frequencies, bandwidth and subsequent data rates. (provide an actual number here) This helps with testing and implementation of different event detection algorithms as improvements are made. For this reason, we won't go into much detail about triggering here.

8.1.2 Event detection

The event detection is directly related to the sensitivity in two ways: 1. Detection efficiency dictates the number of events near the end point which contributes as PTS: $N^{\text{placeholder}}$ -power to sensitivity 2. Backgrounds As a counter-point, the efficiency can be increased arbitrarily at the cost of the increase in the BGs, which contributes as PTS: $N^{\text{placeholder}}$ -power to sensitivity Both efficiency and the BG dependence on energy is particularly important and

care has to be taken. 3. Event detection also plays a very important role in the reconstruction of the event properties, in particular the start frequency extraction which directly effects the energy reconstruction and thus impacts the sensitivity

1–2 paragraphs on the discussion of the pileup.

8.2 Track and Event Reconstruction

PTS: Probably don't need these: Subsystem objectives, operational principles, AoA

T&ER with the extraction of the relevant PTS: kinematic parameters for individual events.

The primary requirement of the CRES for nu mass measurement is to extract the energy of electron from the tritium beta decay. The initial kinematic properties of the electron can be parameterized using the position and the momentum vectors, these can be reparameterized into ... $E, \dots$, where ... While the kinetic energy of the electron is of primary interest, other kinematic parameters are not irrelevant. The kinetic energy is related to the observable/measurable quantity, cyclotron frequency through **ensure that the same terms are used anywhere.**:

$$f_c = \frac{eB}{\gamma m} \quad (21)$$

Since the cyclotron frequency of the electron depends also on the B-field, it is important to know the B-field precisely. In particular, given that the no-work trap enabled by the B-field implies that the field magnitude is position dependent and this means that we need to know the exact value of B field along the electron's trajectory and consequently the electron trajectory itself. Elaborate here on the axial motion and B-field correction.

Brief intro to what CRES events look like (would have already been discussed in 3) and what parameters impact various track and event properties; depends very much on how much was already discussed before. (Will wait until the end to write this).

Phase-II event reconstruction; will keep this short mainly for context but will highlight the successful implementation and pave the way to discuss what needs to be improved and what is missing.

Towards LFA-level reconstruction: Discuss what outcomes from the CCA reconstruction are needed to demonstrate that LFA could meet its goals, this is the meat of the section; highlight the need for understanding the systematic effects.

Lessons to be learned from CCA

CCA will help: 1. Identify how well we can reconstruct energy compared to the best-possible values 2. Show us if our best reconstruction algorithms let us achieve the optimum resolution and efficiency and if not help us identify why not 3. Identify how well we can and will need to reconstruct the radial and azimuthal For the first time Event-by-event reconstruction and associated uncertainties This includes pitch angle-based energy correction, Perhaps even radial position-based energy correction. Azimuth reconstruction may also be needed in case

of non-uniformities Radial position reconstruction more important to model the ensemble spectrum and include the associated uncertainties.

Discuss that CCA provides the test ground for the exploratory work on energy reconstruction including ML for reconstruction; ATM I am thinking to mention that we are exploring ML options with some success but won't go into too many technical details.

Scope of CRES event reconstruction for future: Disk to individual event properties; Discuss how we need to improve efficiency and resolution while maintaining low FB rates and highlight how this can be possible by utilizing the complete event information (which was not done in P-II)

8.3 Spectrum Analysis

Intro of SA: SA includes the ensemble-level analysis of the measured energy spectrum. This includes: 1. Combining the energy (and associated uncertainties) provided by the reconstruction into a spectrum 2. Given the detector response, generation of spectrum models 3. Using 1 and 2, extracting the parameters of interest i.e., energy resolution, neutrino mass etc and corresponding uncertainties

P-II SA procedure (from PRC and Christine's thesis)

What will be different for LFA:

1. New MC spectrum generation based on reconstruction-extracted efficiency and resolution grids
2. Used analytical model
3. Identify how to reduce the biggest contributors of uncertainties; in particular scattering [PTS: the biggest uncertainty for P-II analysis](#) plays an important role in spectrum building, identify the ways to improve it.

How of that can we learn from CCA:

1. Validate new Monte Carlo Spectrum generation; identify the complexities and complications and adjust as needed
2. Test the need and viability of using a numerical analysis model (vs analytic) for better precision
3. Understand the impact of scattering on the measured spectrum (via missed tracks); develop an improved strategy for modeling scattering. In particular play with gas densities to extract this information directly from CRES.

9 Site Selection

Primary Author: Brent, w/ IRs from candidate institutions.

An analysis of alternatives by each candidate site. Clear set of criteria for choosing a site. Primary criteria are:

- Available infrastructure, or prospect to establish it
 - Physical space. Especially ceiling clearance, but also floor space, offices...
 - Services. Electrical, computing/networking, plumbing, ventilation...
 - existing tritium license or potential to obtain one
- Institutional Fortitude
 - Size and Stability of your local group and support staff
 - Past record hosting similar experiments
 - Any additional institutional perks (matching funds, in-kind contributions, statements of commitment from Deans...)
- Accessibility
 - How hard is it to travel there?
 - What capacity for out-of-town collaborators? (hotels, grocery stores, night life...)
- Collaborative Environment. Affirmation and evidence that your site (institution and local community) would be welcoming to all Project 8 collaborators.

9.1 Site Subsystem Template

The first part, at this subsection level, should be a brief introduction that summarizes your institution's case and foreshadows what is to be given in more detail in the subsubsections below.

9.1.1 Physical Resources

Describe the physical resources your institution would provide. See the outline above.

9.1.2 Institutional Resources

Describe the non-physical resources (people, funding...) including relevant past experience. See the outline above.

9.1.3 Accessibility

How hard/easy will it be for collaborators to get to and work at your site? See the outline above.

9.1.4 Collaborative Environment

Your guarantee that all Project 8 collaborators will feel welcome and included. See the outline above.

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A Technology Readiness Levels

Primary Author: Brent

Succinct summary of the DOE NNSA TRL guide

A.1 DOE NNSA TRL guide link

<https://3.basecamp.com/3700981/buckets/3107037/uploads/7379915292>

A.2 Guidance from DOE

From Elizabeth Bartosz on Aug 2, 2024:

Brent, you asked about TRLs as mentioned in DOE Order 413.3B. Here is the relevant quote:

Major System Projects, or first-of-a-kind engineering endeavors, must be assessed prior to each CD using the Technology Readiness Assessment and should achieve the following minimum Technology Readiness Level (TRL) scores for each critical technology item or system as determined by an independent review team outside of the project team before that CD can be approved. The higher the TRL at CD-2, the lower the risk to the project. The PME must provide justification to the ESAAB, if pursuing a TRL less than 7, at CD-2, which in turn will notify the CE. The following represents the minimum TRL at each CD:

- CD-1: TRL 4
- CD-2: TRL 7

For Major System Projects where new critical technologies are being deployed, the TRA shall be conducted and the associated Technology Maturation Plan developed prior to CD-2. On those projects where a significant critical technology element modification occurs subsequent to CD-2, conduct another TRA prior to CD-3. It is strongly encouraged for use by the PME for projects with a TPC less than \$750M.

A GAO guide to technology readiness assessments is here:

<https://www.gao.gov/assets/gao-20-48g.pdf>